



Eventlog Reference Guide for UnityCloud

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Affirmed Networks Inc.

A Microsoft Company

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Consul Read Status

Version

1.0

Description

Reading Previously Stored Status from Consul.

Severity

- Info

Auto Clearing

Yes

Clear Message

Consul Read Response:\$1 at:\$2 NF:\$4 is using the :\$3 InstanceID

Trigger Message

Consul Read Response: **{Consul Response}** at: **{Event TimeStamp}** NF: **{Network Function Name}** is using the : **{Info Message}** InstanceID

where:

- Consul Response is the Consul Read HTTP response
- Event TimeStamp is the Event Timestamp
- Info Message is the Info about the InstanceID (static/dynamic/existing)
- Network Function Name is the Network Function Name where:
- Consul Response is the Consul Read HTTP response
- Event TimeStamp is the Event Timestamp
- Info Message is the Info about the InstanceID (static/dynamic/existing)
- Network Function Name is the Network Function Name

Repair Action

Informative Event about the Consul Read Status, No Action Required

Alert Rules

Alert Rule 1

This event will result in raising an alert for every **Consul Read Status** event generation based on realert configuration. The rising of this alert is for each unique value of **severity**, **podName** and **networkFunctionName** fields with realert time counted separately.

Critical Service not Available

Version

1.0

Description

Critical Service is Unavailable

Severity

- Critical

Auto Clearing

Yes

Clear Message

Critical Service :\$1 has become Available (\$4) at :\$2 NF:\$3

Trigger Message

Critical Service : **{Critical Service Name}** is unavailable (**{Service Status}**) at : **{Event TimeStamp}** NF: **{Network Function Name}**

where:

- Critical Service Name is the Critical Service Name
- Event TimeStamp is the Event Timestamp
- Network Function Name is the Network Function Name
- Service Status is the Available/Required where:
- Critical Service Name is the Critical Service Name
- Event TimeStamp is the Event Timestamp
- Network Function Name is the Network Function Name
- Service Status is the Available/Required

Repair Action

Check if all the required critical services are available

Alert Rules

Alert Rule 1

This event will result in raising an alert for every **Critical Service not Available** event generation based on realert configuration. The rising of this alert is for each unique value of **severity**, **criticalServiceName** and **networkFunctionName** fields with realert time counted separately.

In Memory Buffer Full

Version

1.0

Description

This event indicates that in-memory buffer has become full(alarm set) or become half(alarm cleared) for a given buffer ID

Severity

- Critical

Auto Clearing

Yes

Clear Message

On Pod instance \$2 for a buffer with ID = \$1, in-memory buffer has become half, max buffer size in KB = \$3

Trigger Message

On Pod instance **{UniquePodId}** for a buffer with ID = **{BufferId}** , in-memory buffer has become full, max buffer size in KB = **{MaxBufferSize}**

where:

- BufferId is the Buffer Identifier for which the buffering is applicable
- UniquePodId is the Unique identifier for the pod that has generated this alert
- MaxBufferSize is the Maximum size of this buffer in KB where:
- BufferId is the Buffer Identifier for which the buffering is applicable
- UniquePodId is the Unique identifier for the pod that has generated this alert
- MaxBufferSize is the Maximum size of this buffer in KB

Potential Impact

Some PDUs being dropped from buffer with ID = \$1

Repair Action

Check if there is memory available on the pod and increase the size of the buffer

Alert Rules

Alert Rule 1

This event will result in raising an alert for every **In Memory Buffer Full** event generation based on realert configuration. The rising of this alert is for each unique value of **bufferid**, **uniquepodid** and **maxbuffersize** fields with realert time counted separately.

NF Management Status

Version

1.0

Description

NF Status towards NRF

Severity

- Info

Auto Clearing

Yes

Clear Message

NF:\$4 has updated NRF with status:\$1 at:\$2 InstanceID:\$3

Trigger Message

NF: **{Network Function Name}** has updated NRF with status: **{NfMgmt Status}** at: **{Event TimeStamp}**
InstanceID: **{NF Instance ID}**

where:

- NfMgmt Status is the Nf Management Interface Status(Registered/DeRegistered/Maintenance)
- Event TimeStamp is the Event Timestamp
- NF Instance ID is the Network Function Instance ID
- Network Function Name is the Network Function Name where:

- NfMgmt Status is the Nf Management Interface Status(Registered/DeRegistered/Maintenance)
- Event TimeStamp is the Event Timestamp
- NF Instance ID is the Network Function Instance ID
- Network Function Name is the Network Function Name

Repair Action

Informative Event about the NF Status, No Action Required

Alert Rules

Alert Rule 1

This event will result in raising an alert for every **NF Management Status** event generation based on realert configuration. The rising of this alert is for each unique value of **networkFunctionName**, *** with realert time counted separately.

Unstable NF due to Frequent Restart

Version

1.0

Description

Network Function is unstable due to Frequent Restarts

Severity

- Critical

Auto Clearing

Yes

Clear Message

Critical Service:\$1 is stable.

Trigger Message

Due to Critical Service: **{Critical Service Name}** , NF: **{Network Function Name}** is unstable, timestamp: **{Event TimeStamp}**

where:

- Critical Service Name is the Critical Service Name
- Event TimeStamp is the Event Timestamp
- Network Function Name is the Network Function Name where:

- Critical Service Name is the Critical Service Name

Repair Action

Check why Critical Service instances are restarting

Alert Rules

Alert Rule 1

This event will result in raising an alert for every **Unstable NF due to Frequent Restart** event generation based on realert configuration. The rising of this alert is for each unique value of **severity**, **criticalServiceName** and **networkFunctionName** fields with realert time counted separately.

Unstable NF due to Readiness Failure

Version

1.0

Description

Network Function is unstable due to Readiness Failure

Severity

- Critical

Auto Clearing

Yes

Clear Message

Critical Service:\$1 is stable.

Trigger Message

Due to Critical Service: **{Critical Service Name}** , NF: **{Network Function Name}** is unstable, timestamp: **{Event TimeStamp}**

where:

- Critical Service Name is the Critical Service Name
- Event TimeStamp is the Event Timestamp
- Network Function Name is the Network Function Name where:
- Critical Service Name is the Critical Service Name

Repair Action

Check why there are Readiness Probe failures for the Critical Service

Alert Rules

Alert Rule 1

This event will result in raising an alert for every **Unstable NF due to Readiness Failure** event generation based on realert configuration. The rising of this alert is for each unique value of **severity**, **criticalServiceName** and **networkFunctionName** fields with realert time counted separately.

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Consul Read Status

Version

1.0

Description

Reading Previously Stored Status from Consul.

Severity

- Info

Auto Clearing

Yes

Clear Message

Consul Read Response:\$1 at:\$2 NF:\$4 is using the :\$3 InstanceID

Trigger Message

Consul Read Response: **{Consul Response}** at: **{Event TimeStamp}** NF: **{Network Function Name}** is using the : **{Info Message}** InstanceID

where:

- Consul Response is the Consul Read HTTP response
- Event TimeStamp is the Event Timestamp
- Info Message is the Info about the InstanceID (static/dynamic/existing)
- Network Function Name is the Network Function Name where:
- Consul Response is the Consul Read HTTP response
- Event TimeStamp is the Event Timestamp
- Info Message is the Info about the InstanceID (static/dynamic/existing)
- Network Function Name is the Network Function Name

Repair Action

Informative Event about the Consul Read Status, No Action Required

Alert Rules

Alert Rule 1

This event will result in raising an alert for every **Consul Read Status** event generation based on realert configuration. The rising of this alert is for each unique value of **severity**, **podName** and **networkFunctionName** fields with realert time counted separately.

Critical Service not Available

Version

1.0

Description

Critical Service is Unavailable

Severity

- Critical

Auto Clearing

Yes

Clear Message

Critical Service :\$1 has become Available (\$4) at :\$2 NF:\$3

Trigger Message

Critical Service : **{Critical Service Name}** is unavailable (**{Service Status}**) at : **{Event TimeStamp}** NF: **{Network Function Name}**

where:

- Critical Service Name is the Critical Service Name
- Event TimeStamp is the Event Timestamp
- Network Function Name is the Network Function Name
- Service Status is the Available/Required where:
- Critical Service Name is the Critical Service Name
- Event TimeStamp is the Event Timestamp
- Network Function Name is the Network Function Name
- Service Status is the Available/Required

Repair Action

Check if all the required critical services are available

Alert Rules

Alert Rule 1

This event will result in raising an alert for every **Critical Service not Available** event generation based on realert configuration. The rising of this alert is for each unique value of **severity**, **criticalServiceName** and **networkFunctionName** fields with realert time counted separately.

License expired

Version

1.0

Description

License expired.

Severity

- Info
- Minor
- Major
- Critical

Auto Clearing

Yes

Clear Message

License :\$1 expiration date :\$2

Trigger Message

License : **{License name version}** expiration date : **{License expiration date}**

where:

- License name version is the License name and version.
- License expiration date is the License expiration date. where:
- License name version is the License name and version.
- License expiration date is the License expiration date.

Repair Action

Check license expiration date.

License limit reached

Version

1.0

Description

License limit reachead.

Severity

- Info
- Minor
- Major
- Critical

Auto Clearing

Yes

Clear Message

License :\$1 limit cleared : number of SAUs \$4, entitlements used \$3, entitlements requested \$2

Trigger Message

License : **{License name version}** limit reached : number of SAUs **{Number of SAUs}** , entitlements used **{Number of entitlements requested}** , entitlements requested **{Number of entitlements used}**

where:

- License name version is the License name and version.
- Number of entitlements used is the Number of entitlements used.
- Number of entitlements requested is the Number of entitlements requested.
- Number of SAUs is the Number of SAUs. where:
- License name version is the License name and version.
- Number of entitlements used is the Number of entitlements used.
- Number of entitlements requested is the Number of entitlements requested.
- Number of SAUs is the Number of SAUs.

Repair Action

Check license soft limit.

License required

Version

1.0

Description

License is required.

Severity

- Critical
- Info

Auto Clearing

Yes

Clear Message

License :\$1 is obtained.

Trigger Message

License : **{License name version}** is required

where:

- License name version is the License name and version. where:
 - License name version is the License name and version.
-

License server is not reachable

Version

1.0

Description

License server is not reachables.

Severity

- Critical
- Info

Auto Clearing

Yes

Clear Message

Connected to license server :\$1

Trigger Message

Failed to connect to license server : **{License server}**

where:

- License server is the License server name or ip address. where:
- License server is the License server name or ip address.

Potential Impact

Licenses cannot be obtained from the server.

Repair Action

Check license server connection.

NF Management Status

Version

1.0

Description

NF Status towards NRF

Severity

- Info

Auto Clearing

Yes

Clear Message

NF:\$4 has updated NRF with status:\$1 at:\$2 InstanceID:\$3

Trigger Message

NF: **{Network Function Name}** has updated NRF with status: **{NfMgmt Status}** at: **{Event TimeStamp}**
InstanceID: **{NF Instance ID}**

where:

- NfMgmt Status is the Nf Management Interface Status(Registered/DeRegistered/Maintenance)
- Event TimeStamp is the Event Timestamp
- NF Instance ID is the Network Function Instance ID
- Network Function Name is the Network Function Name where:
- NfMgmt Status is the Nf Management Interface Status(Registered/DeRegistered/Maintenance)
- Event TimeStamp is the Event Timestamp
- NF Instance ID is the Network Function Instance ID
- Network Function Name is the Network Function Name

Repair Action

Informative Event about the NF Status, No Action Required

Alert Rules

Alert Rule 1

This event will result in raising an alert for every **NF Management Status** event generation based on realert configuration. The rising of this alert is for each unique value of **networkFunctionName**, *** with realert time counted separately.

Unstable NF due to Frequent Restart

Version

1.0

Description

Network Function is unstable due to Frequent Restarts

Severity

- Critical

Auto Clearing

Yes

Clear Message

Critical Service:\$1 is stable.

Trigger Message

Due to Critical Service: **{Critical Service Name}** , NF: **{Network Function Name}** is unstable, timestamp: **{Event TimeStamp}**

where:

- Critical Service Name is the Critical Service Name
- Event TimeStamp is the Event Timestamp
- Network Function Name is the Network Function Name where:
- Critical Service Name is the Critical Service Name

Repair Action

Check why Critical Service instances are restarting

Alert Rules

Alert Rule 1

This event will result in raising an alert for every **Unstable NF due to Frequent Restart** event generation based on realert configuration. The rising of this alert is for each unique value of **severity**, **criticalServiceName** and **networkFunctionName** fields with realert time counted separately.

Unstable NF due to Readiness Failure

Version

1.0

Description

Network Function is unstable due to Readiness Failure

Severity

- Critical

Auto Clearing

Yes

Clear Message

Critical Service:\$1 is stable.

Trigger Message

Due to Critical Service: **{Critical Service Name}** , NF: **{Network Function Name}** is unstable, timestamp: **{Event TimeStamp}**

where:

- Critical Service Name is the Critical Service Name
- Event TimeStamp is the Event Timestamp
- Network Function Name is the Network Function Name where:
- Critical Service Name is the Critical Service Name

Repair Action

Check why there are Readiness Probe failures for the Critical Service

Alert Rules

Alert Rule 1

This event will result in raising an alert for every **Unstable NF due to Readiness Failure** event generation based on realert configuration. The rising of this alert is for each unique value of **severity**, **criticalServiceName** and **networkFunctionName** fields with realert time counted separately.

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AMF Config Update Failure

Version

1.0

Description

This event is set when the gNB can not accept AMF Configuration Update from AMF. This event is not automatically cleared and must be manually cleared by an operator.

Severity

- Warning

Auto Clearing

Yes

Clear Message

n/a

Trigger Message

The AMF received an AMF Configuration Update Failure message from the gNB with with Global RAN Node ID: **{Global RAN Node ID}** , gNB Name: **{gNB Name}** and cause code **{Cause}**

where:

- Global RAN Node ID is the gNB identifier in the form of a Global RAN Node ID. This is comprised of PLMN and a unique RAN instance ID.
- gNB Name is the Name of the gNB, or 'not-available' if not provided by the gNB
- Cause is the Cause value from the AMF Configuration Update Failure message

Potential Impact

The AMF Config Update procedure has failed. Failure to communicate the AMF configuration changes to gNB could result in load imbalance among TNLAs and UE procedure failure if served GUAMI list and/or supported PLMN list changes are not communicated to the gNB.

Repair Action

Check with the gNB vendor to determine the meaning of the cause code. Retry the gNB connection after any adjustments are made.

Resource Xpath

Global RAN Node ID and gNB Name

The format of resource xpath is:

```
/amf/gnb/[ {Global RAN Node ID} ]/[ {gNB Name} ]
```

where:

- Global RAN Node ID is the gNB identifier in the form of a Global RAN Node ID. This is comprised of PLMN and a unique RAN instance ID.
- gNB Name is the Name of the gNB, or 'not-available' if not provided by the gNB

Alert Rules

Alert Rule 1

This event will result in raising an alert for every **AMF Config Update Failure** event generation based on realert configuration. The rising of this alert is for each unique value of **globalRanNodeId**, *** with realert time counted separately.

AMF Config Update send failed

Version

1.0

Description

This event indicates that AMF failed to send an AMF Configuration Update message to a gNB. This event is not automatically cleared and must be manually cleared by an operator.

Severity

- Warning

Auto Clearing

Yes

Clear Message

n/a

Trigger Message

AMF failed to send AMF Configuration Update to gNB with Global RAN Node ID: **{Global RAN Node ID}** , gNB Name: **{gNB Name}** , The AMF configuration update is triggered by event: **{Trigger}** .

where:

- Global RAN Node ID is the gNB identifier in the form of a Global RAN Node ID. This is comprised of PLMN and a unique RAN instance ID.
- gNB Name is the Name of the gNB, or 'not-available' if not provided by the gNB.
- Trigger is the Trigger reason for AMF Configuration Update. Possible triggers are: NG Setup, RAN Config Update, AMF Config Update, Resend due to TNLA down.

Potential Impact

AMF configuration changes are not communicated to the gNB. This can be due to socket busy or other network issue. Failure to communicate the AMF configuration changes to gNB could result in load imbalance among TNLAs and UE procedure failure if served GUAMI list and/or supported PLMN list changes are not communicated to the gNB.

Repair Action

The AMF does retry sending the configuration update message, but this event indicates all retries are exhausted. Attempt to reconnect the gNB.

Resource Xpath

Global RAN Node ID and gNB Name

The format of resource xpath is:

```
/amf/gnb/[ {Global RAN Node ID} ]/[ {gNB Name} ]
```

where:

- Global RAN Node ID is the gNB identifier in the form of a Global RAN Node ID. This is comprised of PLMN and a unique RAN instance ID.
- gNB Name is the Name of the gNB, or 'not-available' if not provided by the gNB.

Alert Rules

Alert Rule 1

This event will result in raising an alert for every **AMF Config Update send failed** event generation based on realert configuration. The rising of this alert is for each unique value of **globalRanNodeId**, *** with realert time counted separately.

AMF Config Update success with failed to setup list

Version

1.0

Description

This event is set during AMF Configuration Update Procedure when the gNB sends the AMF Configuration Update Acknowledgement message with AMF TNL Association Failed to Setup List. This event is not automatically cleared and must be manually cleared by an operator.

Severity

- Warning

Auto Clearing

No

Trigger Message

gNB RAN Node ID: **{Global RAN Node ID}** , gNB Name: **{gNB Name}** , failed to establish AMF TNL Associations connection to : **{Failed to set up list}**

where:

- Global RAN Node ID is the gNB identifier in the form of a Global RAN Node ID. This is comprised of PLMN and a unique RAN instance ID.
- gNB Name is the Name of the gNB, or 'not-available' if not provided by the gNB.
- Failed to set up list is the A list of AMF TNL associations that failed to be established.

Potential Impact

This is part of the AMF configuration update procedure initiated by AMF that not all TNL are necessarily established. It can be due to limitations on the gNB, or the network. But failure to establish all TNL could affect the reliability of the gNB connection and the performance of the system. For instance, it affects reliability of the gNB connection as multiple TNL can provide path diversity, and it can affect performance when some TNL are failed as load balancing is limited.

Repair Action

Check the IP and cause code for the failed TNL mentioned in the log. Check with the gNB vendor to determine the meaning of the cause code and whether the gNB supports multiple TNL to AMF. Retry the gNB connection after any adjustments are made.

Resource Xpath

Global RAN Node ID and gNB Name

The format of resource xpath is:

```
/amf/gnb/[ {Global RAN Node ID} ]/[ {gNB Name} ]
```

where:

- Global RAN Node ID is the gNB identifier in the form of a Global RAN Node ID. This is comprised of PLMN and a unique RAN instance ID.
- gNB Name is the Name of the gNB, or 'not-available' if not provided by the gNB.

Alert Rules

Alert Rule 1

This event will result in raising an alert for every **AMF Config Update success with failed to setup list** event generation based on realert configuration. The rising of this alert is for each unique value of **globalRanNodeId**, *** with realert time counted separately.

Consul Read Status

Version

1.0

Description

Reading Previously Stored Status from Consul.

Severity

- Info

Auto Clearing

Yes

Clear Message

Consul Read Response:\$1 at:\$2 NF:\$4 is using the :\$3 InstanceID

Trigger Message

Consul Read Response: **{Consul Response}** at: **{Event TimeStamp}** NF: **{Network Function Name}** is using the : **{Info Message}** InstanceID

where:

- Consul Response is the Consul Read HTTP response
- Event TimeStamp is the Event Timestamp
- Info Message is the Info about the InstanceID (static/dynamic/existing)
- Network Function Name is the Network Function Name where:
- Consul Response is the Consul Read HTTP response
- Event TimeStamp is the Event Timestamp
- Info Message is the Info about the InstanceID (static/dynamic/existing)
- Network Function Name is the Network Function Name

Repair Action

Informative Event about the Consul Read Status, No Action Required

Alert Rules

Alert Rule 1

This event will result in raising an alert for every **Consul Read Status** event generation based on realert configuration. The rising of this alert is for each unique value of **severity**, **podName** and **networkFunctionName** fields with realert time counted separately.

Critical Service not Available

Version

1.0

Description

Critical Service is Unavailable

Severity

- Critical

Auto Clearing

Yes

Clear Message

Critical Service :\$1 has become Available (\$4) at :\$2 NF:\$3

Trigger Message

Critical Service : **{Critical Service Name}** is unavailable (**{Service Status}**) at : **{Event TimeStamp}** NF: **{Network Function Name}**

where:

- Critical Service Name is the Critical Service Name
- Event TimeStamp is the Event Timestamp
- Network Function Name is the Network Function Name
- Service Status is the Available/Required where:
- Critical Service Name is the Critical Service Name
- Event TimeStamp is the Event Timestamp
- Network Function Name is the Network Function Name
- Service Status is the Available/Required

Repair Action

Check if all the required critical services are available

Alert Rules

Alert Rule 1

This event will result in raising an alert for every **Critical Service not Available** event generation based on realert configuration. The rising of this alert is for each unique value of **severity**, **criticalServiceName** and **networkFunctionName** fields with realert time counted separately.

NF Management Status

Version

1.0

Description

NF Status towards NRF

Severity

- Info

Auto Clearing

Yes

Clear Message

NF:\$4 has updated NRF with status:\$1 at:\$2 InstanceID:\$3

Trigger Message

NF: **{Network Function Name}** has updated NRF with status: **{NfMgmt Status}** at: **{Event TimeStamp}**
InstanceID: **{NF Instance ID}**

where:

- NfMgmt Status is the Nf Management Interface Status(Registered/DeRegistered/Maintenance)
- Event TimeStamp is the Event Timestamp
- NF Instance ID is the Network Function Instance ID
- Network Function Name is the Network Function Name where:
- NfMgmt Status is the Nf Management Interface Status(Registered/DeRegistered/Maintenance)
- Event TimeStamp is the Event Timestamp
- NF Instance ID is the Network Function Instance ID
- Network Function Name is the Network Function Name

Repair Action

Informative Event about the NF Status, No Action Required

Alert Rules

Alert Rule 1

This event will result in raising an alert for every **NF Management Status** event generation based on realert configuration. The rising of this alert is for each unique value of **networkFunctionName**, *** with realert time counted separately.

Unstable NF due to Frequent Restart

Version

1.0

Description

Network Function is unstable due to Frequent Restarts

Severity

- Critical

Auto Clearing

Yes

Clear Message

Critical Service:\$1 is stable.

Trigger Message

Due to Critical Service: **{Critical Service Name}** , NF: **{Network Function Name}** is unstable, timestamp: **{Event TimeStamp}**

where:

- Critical Service Name is the Critical Service Name
- Event TimeStamp is the Event Timestamp
- Network Function Name is the Network Function Name where:
- Critical Service Name is the Critical Service Name

Repair Action

Check why Critical Service instances are restarting

Alert Rules

Alert Rule 1

This event will result in raising an alert for every **Unstable NF due to Frequent Restart** event generation based on realert configuration. The rising of this alert is for each unique value of **severity**, **criticalServiceName** and **networkFunctionName** fields with realert time counted separately.

Unstable NF due to Readiness Failure

Version

1.0

Description

Network Function is unstable due to Readiness Failure

Severity

- Critical

Auto Clearing

Yes

Clear Message

Critical Service:\$1 is stable.

Trigger Message

Due to Critical Service: **{Critical Service Name}** , NF: **{Network Function Name}** is unstable, timestamp: **{Event TimeStamp}**

where:

- Critical Service Name is the Critical Service Name
- Event TimeStamp is the Event Timestamp
- Network Function Name is the Network Function Name where:
- Critical Service Name is the Critical Service Name

Repair Action

Check why there are Readiness Probe failures for the Critical Service

Alert Rules

Alert Rule 1

This event will result in raising an alert for every **Unstable NF due to Readiness Failure** event generation based on realert configuration. The rising of this alert is for each unique value of **severity**, **criticalServiceName** and **networkFunctionName** fields with realert time counted separately.

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gNB Reachability

Version

1.0

Description

Transport Network Layer Association (TNLA) status to a gNB has changed. This event is set approximately 10 seconds after the last TNLA to the gNB goes down. The alarm information indicates the remote IP and port of the primary TNLA, which might not have been the last one connected.

It is automatically cleared when at least one TNLA for the gNB is up. Manually clear the alarm if the gNB is not expected to return to service or is removed from the network.

Severity

- Major

Auto Clearing

Yes

Clear Message

Connectivity is up to gNB Global RAN Node ID: \$1, gNB Name: \$2; TNLA remote IP \$3 and port \$4 has connected.

Trigger Message

Connectivity is down to gNB Global RAN Node ID: **{Global RAN Node ID}** , gNB Name: **{gNB Name}** ; TNLA remote IP **{TNLA Remote IP Address}** and port **{TNLA Remote Port}** has gone down.

where:

- Global RAN Node ID is the gNB identifier in the form of a Global RAN Node ID. This is comprised of PLMN and a unique RAN instance ID.
- gNB Name is the Name of the gNB, or 'not-available' if not provided by the gNB.
- TNLA Remote IP Address is the There can be multiple TNLAs to a gNB. When the alert is raised, this is the remote IP of the primary TNLA to the gNB. When the alert is cleared, this is the remote IP of the first TNLA to the gNB, usually the primary.
- TNLA Remote Port is the There can be multiple TNLAs to a gNB. When the alert is raised, this is the remote port of the primary TNLA to the gNB. When the alert is cleared, this is the remote port of the first TNLA to the gNB, usually the primary. where:
- Global RAN Node ID is the gNB identifier in the form of a Global RAN Node ID. This is comprised of PLMN and a unique RAN instance ID.
- gNB Name is the Name of the gNB, or 'not-available' if not provided by the gNB.
- TNLA Remote IP Address is the There can be multiple TNLAs to a gNB. When the alert is raised, this is the remote IP of the primary TNLA to the gNB. When the alert is cleared, this is the remote IP of the first TNLA to the gNB, usually the primary.

- TNLA Remote Port is the There can be multiple TNLAs to a gNB. When the alert is raised, this is the remote port of the primary TNLA to the gNB. When the alert is cleared, this is the remote port of the first TNLA to the gNB, usually the primary.

Potential Impact

Loss of service in the area covered by this gNB.

Repair Action

Verify the health of the gNB and IP network.

Resource Xpath

Global RAN Node ID and gNB Name

The format of resource xpath is:

```
/amf/gnb/[ {Global RAN Node ID} ]/[ {gNB Name} ]
```

where:

- Global RAN Node ID is the gNB identifier in the form of a Global RAN Node ID. This is comprised of PLMN and a unique RAN instance ID.
- gNB Name is the Name of the gNB, or 'not-available' if not provided by the gNB.

Alert Rules

Alert Rule 1

This event will result in raising an alert for every **gNB Reachability** event generation based on realert configuration. The rising of this alert is for each unique value of **severity** and **globalRanNodeId** fields with realert time counted separately.

Table of Contents

- [CDR File Open Error](#)
 - [Disk quota Threshold Reached](#)
 - [Missing CDR File Rotation Profile Configuration](#)
 - [Missing CDR Profile Configuration](#)
 - [Timer Service Unavailable](#)
-

CDR File Open Error

Version

1.0

Description

This event indicates that the system failed to open the CDR file for writing the CDRs.

This could happen due to either the required disk is not available or the disk space is full.

If the disk space is full, either the disk needs to be monitored and archive the files off the disk or configure purge policy to free up the disk space accordingly.

Severity

- Major

Auto Clearing

No

Trigger Message

File open failed for Federation: **{federation}** , recordType: **{record_type}** , fileFormat: **{file_format}**

where:

- federation is the Name of the federation under which CDR generation failed
- record_type is the Type of record, i.e. CHfRecord/PGwRecord/etc
- file_format is the File format, i.e. ASN.1/CSV/etc

Potential Impact

Loss of CDRs

Repair Action

Ensure that the required disk is available with sufficient space for CDR generation

Disk quota Threshold Reached

Version

1.0

Description

This event indicates that the disk size threshold configured through CDRAgent profile has reached.

If not cleared, this could result in loss of CDRs due to lack of disk space availability.

Ensure that appropriate CDR purge policy is configured to free up some disk space or pull the CDRs out to a safe location.

Severity

- Critical

Auto Clearing

Yes

Clear Message

Disk Space is now available for CDR generation

Trigger Message

The disk space utilization for the directory **{directory}** has crossed the configured high threshold of **{high_threshold}** percentage - Alarm will be cleared when the usage drops below the configured low threshold of **{low_threshold}** .

where:

- high_threshold is the High threshold configured for disk usage, beyond which alarm will be raised
- low_threshold is the Low threshold configured for disk usage, below which alarm will be cleared
- directory is the Mounted directory path

Potential Impact

Failure in CDRs generation

Repair Action

Operator should either monitor their disk usage and archive files off the disk or configure purge policy to free up disk space accordingly.

Missing CDR File Rotation Profile Configuration

Version

1.0

Description

This event indicates that the identified CDR file rotation profile, which is supposed to be used for CDR generation, is missing.

The received CDR will be dropped due to this missing configuration

Severity

- Major

Auto Clearing

No

Trigger Message

Missing CDR File Rotation Profile Name: ***{file_rotation_profile}*** for federation: ***{federation}*** , recordType: ***{record_type}*** , fileFormat: ***{file_format}***

where:

- file_rotation_profile is the Name of the CDR file rotation profile identified for CDR generation
- federation is the Name of the federation under which CDR generation failed
- record_type is the Type of record, i.e. CHfRecord/PGwRecord/etc
- file_format is the File format, i.e. ASN.1/CSV/etc

Potential Impact

Loss of CDRs

Repair Action

Ensure that the system is configured with correct CDR file rotation profiles

Missing CDR Profile Configuration

Version

1.0

Description

This event indicates that the received CDR profile, which is supposed to be used for CDR generation, is missing.

The received CDR will be dropped due to this missing configuration

Severity

- Major

Auto Clearing

No

Trigger Message

Missing CDR Profile Name: ***{cdr_profile}*** for federation: ***{federation}*** , recordType: ***{record_type}*** , fileFormat: ***{file_format}***

where:

- cdr_profile is the Name of the CDR profile received for CDR generation
- federation is the Name of the federation under which CDR generation failed
- record_type is the Type of record, i.e. CHfRecord/PGwRecord/etc
- file_format is the File format, i.e. ASN.1/CSV/etc

Potential Impact

Loss of CDRs

Repair Action

Ensure that the system is configured with correct CDR profiles

Timer Service Unavailable

Version

1.0

Description

This event indicates that timer service is unavailable.

If not cleared, this could result in open orphan CDR files of dead POD can be left out as unclaimed in open directory.

Ensure that timer service is available for use.

Severity

- Warning

Auto Clearing

Yes

Clear Message

Timer service is available for use

Trigger Message

Unable to start timer required for claiming open orphan CDR files of dead POD

Potential Impact

Unclaimed open orphan cdr files of dead POD can be left out in open directory

Repair Action

Operator should make sure timer service is deployed for the usage.

Table of Contents

- [Hashicorp Vault not reachable error](#)
 - [Hashicorp Vault not reachable error during certificate renewal](#)
-

Hashicorp Vault not reachable error

Version

1.0

Description

This event alert indicates that Hashicorp vault is not reachable

Severity

- Major

Auto Clearing

No

Root Cause

Hashicorp Vault is not reachable

Trigger Message

Failed to initialise Vault

Potential Impact

Until Hashicorp Vault is reachable no requests can be processed. Cert manager will keep retrying, once vault is reachable requests will be handled automatically.

Repair Action

Please check the status of Vault and verify the connectivity between Vault and the kubernetes cluster which has cert_manager deployed.

Alert Rules

Alert Rule 1

This event will result in raising an alert for every **Hashicorp Vault not reachable error** event generation based on realert configuration.

Hashicorp Vault not reachable error during certificate renewal

Version

1.0

Description

This event alert indicates that Hashicorp vault is not reachable during certificate request

Severity

- Major

Auto Clearing

No

Root Cause

Hashicorp Vault is not reachable

Trigger Message

Cert Manager failed to initialise Hashicorp Vault client for signing as it's not reachable.

Potential Impact

Until Hashicorp Vault is reachable certificate requests can't be processed. Cert manager will keep retrying, once vault is reachable certificate request will be handled automatically.

Repair Action

Please check the status of Vault and verify the connectivity between Vault and the kubernetes cluster which has cert_manager deployed.

Alert Rules

Alert Rule 1

This event will result in raising an alert for every **Hashicorp Vault not reachable error during certificate renewal** event generation based on realert configuration.

Table of Contents

-
- [Configuration Schema Removal Status](#)
 - [Configuration Schema Upgrade Status](#)
 - [Configuration Transaction Commit Event](#)
 - [Disk Access Status](#)
-

Configuration Schema Removal Status

Version

1.0

Description

Configuration schema removal status.

Severity

- Minor
- Info

Auto Clearing

Yes

Clear Message

Configuration schema was successfully deleted. Service name :\$1

Trigger Message

Configuration schema deletion failed. Service name : **{ServiceName}** ; diagnostics [**{Diagnostic}**]

where:

- ServiceName is the Configuration schema service name.
- Diagnostic is the Diagnostic (if any) that may help point to root cause. where:
- ServiceName is the Configuration schema service name.

Potential Impact

Old config schema is still visible in configuraton manager.

Repair Action

Fix errors based on diagnostic and retry schema removal command.

Configuration Schema Upgrade Status

Version

1.0

Description

Configuration schema was upgrade status, triggered when a new version/type of pod appears in the system.

Severity

- Major
- Info

Auto Clearing

Yes

Clear Message

Configuration schema was successfully upgraded. Service name :\$1

Trigger Message

Configuration schema upgrade failed. Service name : **{ServiceName}** , diagnostics [**{Diagnostic}**]

where:

- ServiceName is the Configuration schema service name.
- Diagnostic is the Diagnostic (if any) that may help point to root cause. where:
- ServiceName is the Configuration schema service name.

Potential Impact

Configuration manager pod is not upgraded with latest schema changes for given service.

Repair Action

Fix errors based on diagnostic and reset affected pods.

Configuration Transaction Commit Event

Version

1.0

Description

Configuration transaction was successfully committed.

Severity

- Info

Auto Clearing

Yes

Clear Message

n/a

Trigger Message

Configuration transaction committed successfully, transactionId: **{Configuration transcation id}**

where:

- Configuration transcation id is the Configuration transaction id.

Disk Access Status

Version

1.0

Description

Disk access status.

Severity

- Critical

Auto Clearing

Yes

Clear Message

n/a

Trigger Message

Failed to save file with : **{Reason}**

where:

- Reason is the Reason for failure.

Potential Impact

Configuration modifications are not reaching applications.

Repair Action

Fix disk access errors based on given reason and reset application pods that could be affected.

Table of Contents

- [CHF Server Registration Failure with Egress Gateway](#)
 - [CHF Server Registration with Egress Gateway Status](#)
 - [Charging Function Server Association Status](#)
 - [Kafka Broker Connection Status](#)
 - [Local CDR Agent Association Status](#)
 - [Missing Charging Function Profile Configuration](#)
 - [Nchf Unaccounted Usage Report Event](#)
-

CHF Server Registration Failure with Egress Gateway

Version

1.0

Description

This event indicates the registration failure of a given CHF server with the Egress Gateway.

Severity

- Major

Auto Clearing

No

Trigger Message

CHF Server **{peerDetails}** Registration failure.

where:

- peerDetails is the CHF Server details

Potential Impact

Failure in registration of CHF server with Egress Gateway impacts the communication with the server which in turn impacts the charging of the subscribers.

Repair Action

Ensure the istio-config agent is available for external hosts registration.

Alert Rules

Alert Rule 1

This event will result in raising an alert for every **CHF Server Registration Failure with Egress Gateway** event generation based on realert configuration.

CHF Server Registration with Egress Gateway Status

Version

1.0

Description

This event indicates the status of registration of a given CHF server with the Egress Gateway to enable the communication with it.

Severity

- Major

Auto Clearing

No

Trigger Message

CHF Server FQDN **{fqdn}** address resolution failed.

where:

- fqdn is the FQDN of CHF Server

Potential Impact

Failure in registration of CHF server with Egress Gateway impacts the communication with the server which in turn impacts the charging of the subscribers.

Repair Action

Ensure either the locally configured CHF server FQDN is locally resolvable to identify the IP address of it or explicitly configure the IP Address of the server along with the FQDN in the CHF server profile configured.

Alert Rules

Alert Rule 1

This event will result in raising an alert for every **CHF Server Registration with Egress Gateway Status** event generation based on realert configuration.

Charging Function Server Association Status

Version

1.0

Description

This event indicates the association state between ChfAgent and external Charging Function(CHF)

Severity

- Major

Auto Clearing

Yes

Clear Message

Association to Charging Function (CHF) \$1 has recovered

Trigger Message

Association to Charging Function (CHF) **{apiRoot}** has gone down

where:

- apiRoot is the Address of the Charging Function(CHF) where:
- apiRoot is the Address of the Charging Function(CHF)

Potential Impact

Failure in establishing charging session or loss of reporting usage to Charging Function

Alert Rules

Alert Rule 1

This event will result in raising an alert when **currentState** field match with any of the disallowed values.

Disallowed Values:

- **DOWN**

Alert Rule 2

This event will result in raising an alert when **currentState** field are not set with any of the allowed values.

Allowed Values:

- **DOWN**

Kafka Broker Connection Status

Version

1.0

Description

This event is generated when the connection status towards Kafka broker changes. A critical event is raised when the Kafka broker connection goes down. The event is cleared when the connection is restored.

Severity

- Critical

Auto Clearing

Yes

Clear Message

Connection to Kafka broker \$1 is up

Trigger Message

Connection to Kafka broker **{Kafka Broker IP And Port}** is down

where:

- Kafka Broker IP And Port is the Configured Kafka broker endpoint in the format IP:port or FQDN:port. If port is not provided, a default port number 9092 is assumed. where:
- Kafka Broker IP And Port is the Configured Kafka broker endpoint in the format IP:port or FQDN:port. If port is not provided, a default port number 9092 is assumed.

Potential Impact

Applications will not be able to produce/consume the messages from Kafka Broker. This could have an impact based on the use cases like EDR generation, storing charging records, streaming packets etc.

Repair Action

Make sure the Kafka broker IP and port are configured correctly and the broker is reachable.

Local CDR Agent Association Status

Version

1.0

Description

This alarm indicates the association state between ChfAgent and Local CDR Agent services.

Severity

- Major

Auto Clearing

Yes

Clear Message

Association to Local CDR Agent has recovered

Trigger Message

Association to Local CDR Agent has gone down

Potential Impact

Local CDRs are not generated as Local CDR Agent service is not available

Repair Action

Ensure Local CDR Agent service is up and running

Missing Charging Function Profile Configuration

Version

1.0

Description

This event indicates that the identified Charging Function(CHF) profile configuration through the chfAgent profile is not configured.

This results in failure in identification of CHF server with which charging needs to be done.

Severity

- Major

Auto Clearing

No

Trigger Message

Missing CHF profile configuration: **{*chf_profile*}**

where:

- chf_profile is the Name of the CHF profile configured

Potential Impact

Failure in identification of CHF server

Repair Action

Ensure that the system is configured correctly

Alert Rules

Alert Rule 1

This event will result in raising an alert for every **Missing Charging Function Profile Configuration** event generation based on realert configuration.

Nchf Unaccounted Usage Report Event

Version

1.0

Description

This event indicates the unaccounted usage generation due to CHF server and Kafka Message Broker unavailability because of which the usage will be discarded.

This event is applicable only when the One Time Usage Replay feature is enabled.

Severity

- Major

Auto Clearing

No

Trigger Message

Unaccounted Usage Report: ***{usage_report}*** . Replay Topic Name: ***{replay_topic_name}***

where:

- `usage_report` is the The complete usage report which is supposed to be reported to CHF server but is unaccounted due to unavailability of both CHF server and the Kafka Message Broker
- `replay_topic_name` is the Name of the replay topic

Potential Impact

This event directly indicates the revenue loss due to discarded usage going as unaccounted.

Repair Action

Ensure the Kafka Message Broker is available for replaying the usage once the CHF server is up.

Alert Rules

Alert Rule 1

This event will result in raising an alert for every ***Nchf Unaccounted Usage Report Event*** event generation based on realert configuration.

Table of Contents

- [Application Core Generation Event](#)
 - [Application Core Persistence Failure](#)
-

Application Core Generation Event

Version

1.0

Description

This event indicates that some application running in a given namespace is crashing resulting in generation of ELF core files.

Severity

- Critical

Auto Clearing

No

Trigger Message

{Core File Name} core file got generated for application **{Application Name}** running in container **{Container Name}** in namespace **{Application Namespace}**

where:

- Core File Name is the Name of the core file generated. The format of the core file name is as below:
 - {K8s Namespace}.{POD Name}.{Container Name}.{Application Executable Name}.{Process PID}.{Epoch Timestamp}.core.tar.gz
 - (or)
 - {POD Name}.{Application Executable Name}.{Process PID}.{Epoch Timestamp}.core.tar.gz
- Application Namespace is the Name of the application namespace in which the core file is generated
- Container Name is the Name of the container in which the crashed application was running
- Application Name is the Name of the application which crashed

Potential Impact

Applications crashing could result in service impact as they could be unavailable to provide its service

Repair Action

Identify the crashing application and the root cause of the crash to fix it

Alert Rules

Alert Rule 1

This event will result in raising an alert for every **Application Core Generation Event** event generation based on realert configuration. The rising of this alert is for each unique value of **applicationNamespace**, **containerName** and **applicationName** fields with realert time counted separately.

Application Core Persistence Failure

Version

1.0

Description

This event indicates that the ELF core files generated due to application crashes could not be saved on the provisioned persistent storage.

Severity

- Major

Auto Clearing

No

Trigger Message

Failure in persisting **{Core File Name}** core file generated for an application in namespace **{Application Namespace}**

where:

- Core File Name is the Name of the core file generated
- Application Namespace is the Name of the application namespace in which the core file is generated

Potential Impact

Failure in persisting the ELF core files will result in accumulation of these files on cluster worker nodes resulting in worker node running out of disk space.

Repair Action

Ensure that the provisioned persistent storage is configured correctly or enough disk space is available.

Alert Rules

Alert Rule 1

This event will result in raising an alert for every **Application Core Persistence Failure** event generation based on realert configuration. The rising of this alert is for each unique value of **applicationNamespace**, *** with realert time counted separately.

Table of Contents

- [Diameter Node Min Cfg Status](#)
 - [Diameter Peer Association Status](#)
 - [Etcd Server Association Status](#)
-

Diameter Node Min Cfg Status

Version

1.0

Description

This event indicates the diameter node minimum config status with a diameter peer.

This event is raised when the diameter group minimum configuration is not done.

Once the configuration is done properly, this event will be cleared.

Severity

- Major

Auto Clearing

Yes

Clear Message

Minimum Config for diameter group \$1 is available now

Trigger Message

Minimum Config for diameter group **{group_name}** is not present

where:

- group_name is the Name of the diameter group configured where:
- group_name is the Name of the diameter group configured

Potential Impact

The applications will not be able to communicate with the configured diameter peer which could result in revenue loss.

Repair Action

Ensure that the configured diameter group & peer config is proper so that allowing diameter connections from its peers.

Resource Xpath

Object identifier for the configured diameter group

The format of resource xpath is:

```
diameter-base/diameter-group/ {group_name}
```

where:

- group_name is the Name of the diameter group configured

Diameter Peer Association Status

Version

1.0

Description

This event indicates the association status with a diameter peer.

This event is raised when the diameter connection is lost with the diameter peers for any reason.

Once the connection is established back, this event will be cleared.

Severity

- Major

Auto Clearing

Yes

Clear Message

Connection with diameter peer \$2 of diameter group \$1 is up

Trigger Message

Connection with diameter peer **{peer_name}** of diameter group **{group_name}** is down

where:

- group_name is the Name of the diameter group configured
- peer_name is the Name of the diameter peer configured under a diameter group where:
- group_name is the Name of the diameter group configured
- peer_name is the Name of the diameter peer configured under a diameter group

Potential Impact

The applications will not be able to communicate with the configured diameter peer which could result in revenue loss.

Repair Action

Ensure that the configured diameter peer is up allowing diameter connections from its peers.

Resource Xpath

Object identifier for the configured peer

The format of resource xpath is:

```
diameter-base/diameter-group/ {group_name} /peer/ {peer_name}
```

where:

- group_name is the Name of the diameter group configured
- peer_name is the Name of the diameter peer configured under a diameter group

Etcd Server Association Status

Version

1.0

Description

This event indicates the association status with a etcd server.

This event is raised when the etcd service down notification is received.

Once the etcd server is back up the alarm will be cleared

Severity

- Major

Auto Clearing

Yes

Clear Message

Connection with Etcd service \$1 is up

Trigger Message

Connection with Etcd service **{serviceName}** is down

where:

- serviceName is the Etcd Service Name where:
- serviceName is the Etcd Service Name

Potential Impact

The applications will not be able to communicate with etcd. As a result diameter origin host id allocation will fail

Repair Action

Check the Etcd service. Consider restarting the diameter pod if origin host ID is not allocated even if the alarm is cleared

Resource Xpath

Object identifier for the etcd service

The format of resource xpath is:

```
diameter-base/etcd-service/ {serviceName}
```

where:

- serviceName is the Etcd Service Name

Table of Contents

- [DNS Server Reachability](#)
 - [DNS View Reachability](#)
-

DNS Server Reachability

Version

2.0

Description

A configured DNS Server has changed reachability status.

Severity

- Major

Auto Clearing

Yes

Clear Message

DNS Server \$2/\$3 in view \$1 (domain: '\$4') is reachable or has been removed from configuration

Root Cause

Reachability issue may be related to a problem with the server or with the network.

Trigger Message

DNS Server **{DNS Server Name}** / **{DNS Server IP Address}** in view **{View}** (domain: ' **{Domain}** ') is not reachable

where:

- View is the DNS View where Server is defined
- DNS Server Name is the Name of DNS Server
- DNS Server IP Address is the IP Address of DNS Server
- Domain is the Domain associated with this DNS Server where:
- View is the DNS View where Server is defined
- DNS Server Name is the Name of DNS Server
- DNS Server IP Address is the IP Address of DNS Server
- Domain is the Domain associated with this DNS Server

Potential Impact

Transactions will rely on static-cache first, then check dynamic cache. If stale cache answers are enabled (dns-agent * general stale-answer-enable), a cached answer can be returned up to the configured TTL (dns-agent * general stale-answer-ttl) past their normal expirations. If there is a cache miss, then an external query to a DNS server will be made. This can result in NF functionality failures.

Repair Action

Check the network connectivity between the NF and the DNS server and check the health of the DNS server. If the server or the network has intermittent issues, consider enabling stale cache answers.

Resource Xpath

DNS Server Identifier

The format of resource xpath is:

```
/dns-agent/view[ {View} ]/server[ {DNS Server Name} ]
```

where:

- View is the DNS View where Server is defined
- DNS Server Name is the Name of DNS Server

Alert Rules

Alert Rule 1

This event will result in raising an alert for every **DNS Server Reachability** event generation based on realert configuration. The rising of this alert is for each unique value of **severity**, **view**, **dnsServerName** and **dnsServerIpAddress** fields with realert time counted separately.

DNS View Reachability

Version

2.0

Description

All configured DNS Servers in a view are not reachable.

Severity

- Critical

Auto Clearing

Yes

Clear Message

View \$1 is reachable or server related configuration has changed

Root Cause

Reachability issue may be related to a problem with the server(s) or with the network.

Trigger Message

All configured DNS servers in view **{View}** are not reachable

where:

- View is the DNS View where Server is defined where:
- View is the DNS View where Server is defined

Potential Impact

Transactions will rely on static-cache first, then check dynamic cache. If stale cache answers are enabled (dns-agent * general stale-answer-enable), a cached answer can be returned up to the configured TTL (dns-agent * general stale-answer-ttl) past their normal expirations. If there is a cache miss, then an external query to a DNS server will be made. This can result in NF functionality failures.

Repair Action

Check the network connectivity between the NF and the DNS server(s) and check the health of the DNS server(s). If the server(s) or the network has intermittent issues, consider enabling stale cache answers.

Resource Xpath

DNS View Identifier

The format of resource xpath is:

```
/dns-agent/view[ {View} ]
```

where:

- View is the DNS View where Server is defined

Alert Rules

Alert Rule 1

This event will result in raising an alert for every **DNS View Reachability** event generation based on realert configuration. The rising of this alert is for each unique value of **severity** and **view** fields with realert time counted separately.

Table of Contents

- [BFD Session Status](#)
 - [BGP Session Status](#)
 - [Interface Status](#)
-

BFD Session Status

Version

1.0

Description

This Event is raised when BFD session moves from UP to DOWN state. This Event is cleared when BFD session moves backs to UP state.

Severity

- Major

Root Cause

The connectivity to peer router is no more.

Trigger Message

BFD Session for local address: **{Local Address}** , peer address: **{Peer Address}** is in DOWN state.

where:

- Local Address is the Local address of the BFD session
- Peer Address is the Peer address of the BFD session

Clear Message

BFD Session for local address: **{Local Address}** , peer address: **{Peer Address}** is back to UP state.

where:

- Local Address is the Local address of the BFD session
- Peer Address is the Peer address of the BFD session

Potential Impact

If the BFD session is associated with static routes, the static routes will be deleted or if it is associated with BGP session, the BGP session will go DOWN which will result in deleting all the routes learned via the BGP session.

Repair Action

Verify the connectivity to peer router is UP, check for interface state, ip address.

Resource Xpath

UPF PPE Network-Context BFD Session identifier

The format of resource xpath is:

```
/fed-upf/ {PPE instance} / {Network Context Name} /( {Local Address} , {Peer Address} , {Interface Name} )
```

where:

- Local Address is the Local address of the BFD session
- Peer Address is the Peer address of the BFD session
- Interface Name is the Interface Name on which the BFD session is running
- Network Context Name is the Network Context Name of the BFD session
- PPE instance is the Name of the ppe instance where the BFD Session is present

BGP Session Status

Version

1.0

Description

This Event is raised when BGP session moves from Established state. This Event is cleared when session comes back to Established state.

Severity

- Major

Root Cause

The Event is raised when peering restarts, cleared, or when a peering leaves Established state due to a protocol or networking event.

Trigger Message

BGP Session for local address: **{Local Address}** , peer address: **{Peer Address}** is no more in Established state.

where:

- Local Address is the Local address of the BGP session
- Peer Address is the Peer address of the BGP session

Clear Message

BGP Session for local address: **{Local Address}** , peer address: **{Peer Address}** is back to Established state.

where:

- Local Address is the Local address of the BGP session
- Peer Address is the Peer address of the BGP session

Potential Impact

BGP session is not Established state, it will result in deleting all the routes learned via the BGP session.

Repair Action

Verify the connectivity to peer router is UP, configurations are fine.

Resource Xpath

UPF PPE Network-Context BGP Session identifier

The format of resource xpath is:

```
/fed-upf/ {PPE instance} / {Network Context Name} /( {Local Address} , {Peer Address} )
```

where:

- Local Address is the Local address of the BGP session
- Peer Address is the Peer address of the BGP session
- Network Context Name is the Network Context Name of the BGP session
- PPE instance is the Name of the ppe instance where the BGP Session is present

Interface Status

Version

1.0

Description

This Event is raised when Interface moves to DOWN state. This Event is cleared when Interfaces moves back to UP state

Severity

- Critical

Trigger Message

Interface ***{Interface Name}*** is in DOWN state.

where:

- Interface Name is the Name of the interface

Clear Message

Interface ***{Interface Name}*** is back to UP state

where:

- Interface Name is the Name of the interface

Potential Impact

All the dynamic protocols like BGP, BFD running over this interface will go DOWN. All the traffic going over this interface will be dropped. If there are multiple interfaces, until route convergence happens traffic lost will happen. If there are no other interfaces in the PPE, the PPE is not anymore usable.

Resource Xpath

UPF PPE Network-Context Interface identifier

The format of resource xpath is:

```
/fed-upf/ {PPE instance} / {Network Context Name} / {Interface Name}
```

where:

- Interface Name is the Name of the interface
- Network Context Name is the Network Context Name of the Interface
- PPE instance is the Name of the ppe instance where the interface is present

Table of Contents

- [Elasticsearch CRO status](#)
- [Elasticsearch Index Pressure Event](#)

Elasticsearch CRO status

Version

1.0

Description

This event alert indicates that the elasticsearch CRO status is NotReady.

Severity

- Critical
- Cleared

Auto Clearing

Yes

Clear Message

Elasticsearch CRO status is Provisioning

Trigger Message

Elasticsearch CRO status is NotReady, waiting for it to recover...

Potential Impact

The Elasticsearch may have critically failed and may not function as expected to store or query for the data.

Repair Action

Diagnose the Elasticsearch given the log information. It may be necessary to redeploy or wait for it to recover.

Alert Rules

Alert Rule 1

This event will result in raising an alert due to change with respect to **severity** field based on realert configuration. The rising of this alert is for each unique value of **kubernetes.namespace_name** field with realert time counted separately.

Elasticsearch Index Pressure Event

Version

1.0

Description

This event alert indicates that the elasticsearch had an Indexing pressure, hence memory is adjusted twice.

Severity

- Critical

Auto Clearing

Yes

Clear Message

Elasticsearch Indexing Pressure

Trigger Message

Elasticsearch index pressure is adjusted twice, waiting for it to recover...

Potential Impact

The Elasticsearch may have critically failed and may not function as expected to store or query for the data.

Repair Action

Diagnose the Elasticsearch given the log information. It may be necessary to redeploy or wait for it to recover.

Alert Rules

Alert Rule 1

This event will result in raising an alert for every **Elasticsearch Index Pressure Event** event generation based on realert configuration.

Table of Contents

- [ETCD Server Availability Status](#)
 - [ETCD Server Data Loss Event](#)
-

ETCD Server Availability Status

Version

1.0

Description

This event is generated when the ETCD Server Availability status changes. A critical event is raised when the ETCD Server is not reachable. The event is cleared when the ETCD Server is available.

Severity

- Critical

Auto Clearing

Yes

Clear Message

ETCD Server \$1 is up

Trigger Message

ETCD Server **{ETCD Server FQDN}** is down

where:

- ETCD Server FQDN is the Configured ETCD Server FQDN. where:
- ETCD Server FQDN is the Configured ETCD Server FQDN.

Potential Impact

Applications will not be able to write to or read from ETCD Server.

Repair Action

Make sure the ETCD Server FQDN and port are configured correctly and the ETCD Server is reachable.

Alert Rules

Alert Rule 1

This event will result in raising an alert for every **ETCD Server Availability Status** event generation based on realert configuration. The rising of this alert is for each unique value of **etcdServerFqdn**, *** with realert time counted separately.

ETCD Server Data Loss Event

Version

1.0

Description

This Critical event is raised when there is data loss in ETCD Server. This event should be manually cleared when the ETCD server becomes available after it went down.

Severity

- Critical

Auto Clearing

No

Trigger Message

Data loss in ETCD Server **{ETCD Server FQDN}**

where:

- ETCD Server FQDN is the Configured ETCD Server FQDN.

Potential Impact

Applications that rely on the data stored in ETCD Server will stop functioning temporarily or behave unpredictably.

Repair Action

Make sure the NF virtual restart is triggered

Alert Rules

Alert Rule 1

This event will result in raising an alert for every **ETCD Server Data Loss Event** event generation based on realert configuration. The rising of this alert is for each unique value of **etcdServerFqdn**, *** with realert time counted separately.

Table of Contents

- [ETCD Cluster Creation Failure Event](#)
 - [ETCD Cluster Member Addition Failure Event](#)
 - [ETCD Cluster Setup Failed After Retry Operation Event](#)
 - [ETCD Failure in Replacing Dead Member Event](#)
 - [ETCD Failure to Update CR Status Event](#)
 - [ETCD Retry Operation Failed Event](#)
 - [ETCD Seed Member Creation Failure Event](#)
-

ETCD Cluster Creation Failure Event

Version

1.0

Description

This event alert indicates that the etcd operator has encountered an issue while attempting to create the etcd cluster.

Severity

- Critical

Auto Clearing

No

Root Cause

There can be various reasons which can cause this issue, like Insufficient resources or any issue related to OPA. Actual error which will be part of same error message. Based on the error message we can take next steps.

Trigger Message

cluster failed to setup **{Failure message}**

where:

- Failure message is the Error message for the cluster failure

Potential Impact

Etc-d-operator is unable to create cluster because of which there will be a performance impact.

Repair Action

ETCD should auto-recover, If the etcd cluster doesn't auto recover then collect fed-db-etcd Kargo logs and follow MOP to re-install the federation.

Alert Rules

Alert Rule 1

This event will result in raising an alert for every **ETCD Cluster Creation Failure Event** event generation based on realert configuration.

ETCD Cluster Member Addition Failure Event

Version

1.0

Description

This event alert indicates that the etcd operator has encountered an issue while attempting to add a member to the etcd cluster.

Severity

- Warning

Auto Clearing

No

Root Cause

There can be various reasons which can cause this issue, like Insufficient resources or any issue related to OPA. Actual error which will be part of same error message. Based on the error message we can take next steps.

Trigger Message

add one member failed: creating etcd client failed **{Failure message}**

where:

- Failure message is the Error message for the cluster failure

Potential Impact

The desire number of etcd cluster members will not be available. Hence it could result in failure in forming the quorum making the cluster unavailable for use.

Repair Action

ETCD should auto-recover, If the etcd cluster doesn't auto recover then collect fed-db-etcd Kargo logs and follow MOP to re-install the federation.

Alert Rules

Alert Rule 1

This event will result in raising an alert for every **ETCD Cluster Member Addition Failure Event** event generation based on realert configuration.

ETCD Cluster Setup Failed After Retry Operation Event

Version

1.0

Description

This event alert indicates that the etcd operator has encountered an issue while attempting to create the etcd cluster after retry operation.

Severity

- Critical

Auto Clearing

No

Root Cause

There can be various reasons which can cause this issue, like Insufficient resources or any issue related to OPA. Actual error which will be part of same error message. Based on the error message we can take next steps.

Trigger Message

cluster failed to setup after retry operation ***{Failure message}***

where:

- Failure message is the Error message for the cluster failure

Potential Impact

The etcd cluster will be in bad state and functionality will not be as expected.

Repair Action

N/A

Alert Rules

Alert Rule 1

This event will result in raising an alert for every ***ETCD Cluster Setup Failed After Retry Operation Event*** event generation based on realert configuration.

ETCD Failure in Replacing Dead Member Event

Version

1.0

Description

This event alert indicates that the etcd operator has encountered an issue while attempting to replace the dead member of the etcd cluster.

Severity

- Warning

Auto Clearing

No

Root Cause

There can be various reasons which can cause this issue, like Insufficient resources or any issue related to OPA. Actual error which will be part of same error message. Based on the error message we can take next steps.

Trigger Message

failed to replace dead member ***{Failure message}***

where:

- Failure message is the Error message for the cluster failure

Potential Impact

There will be a performance impact from etcd cluster since one of the member is not active.

Repair Action

N/A

Alert Rules

Alert Rule 1

This event will result in raising an alert for every ***ETCD Failure in Replacing Dead Member Event*** event generation based on realert configuration.

ETCD Failure to Update CR Status Event

Version

1.0

Description

This event alert indicates that the etcd operator has encountered an issue while handling Update operation on etcd cluster.

Severity

- Warning

Auto Clearing

No

Root Cause

There can be various reasons which can cause this issue, like etcd operator might be in bad state or any resource related issues. Actual error which will be part of same error message. Based on the error message we can take next steps.

Trigger Message

failed to update CR status ***{Failure message}***

where:

- Failure message is the Error message for the cluster failure

Potential Impact

There will be a performance impact from etcd cluster since CR status was not updated.

Repair Action

N/A

Alert Rules

Alert Rule 1

This event will result in raising an alert for every ***ETCD Failure to Update CR Status Event*** event generation based on realert configuration.

ETCD Retry Operation Failed Event

Version

1.0

Description

This event alert indicates that the etcd operator has performed retry operation for threshold times and failed to bring up the cluster.

Severity

- Warning

Auto Clearing

No

Root Cause

There can be various reasons which can cause this issue, like Insufficient resources or any issue related to OPA. Actual error which will be part of same error message. Based on the error message we can take next steps.

Trigger Message

Retry operation failed **{Failure message}**

where:

- Failure message is the Error message for the cluster failure

Potential Impact

There will be a performance impact from etcd cluster since the cluster is in bad state.

Repair Action

N/A

Alert Rules

Alert Rule 1

This event will result in raising an alert for every **ETCD Retry Operation Failed Event** event generation based on realert configuration.

ETCD Seed Member Creation Failure Event

Version

1.0

Description

This event alert indicates that the etcd operator has encountered an issue while attempting to create the initial seed member for the etcd cluster.

Severity

- Critical

Auto Clearing

No

Root Cause

There can be various reasons which can cause this issue, like Insufficient resources or any issue related to OPA. Actual error which halted the creation of seed member will be part of same error message. Based on the error message we can take next steps.

Trigger Message

failed to create seed member ***{Failure message}***

where:

- Failure message is the Error message for the cluster failure

Potential Impact

The entire etcd cluster creation process will be halted, since the etcd-operator is unable to create the initial seed member.

Repair Action

N/A

Alert Rules

Alert Rule 1

This event will result in raising an alert for every **ETCD Seed Member Creation Failure Event** event generation based on realert configuration.

Table of Contents

- [Kafka Broker Connection Status](#)
-

Kafka Broker Connection Status

Version

1.0

Description

This event is generated when the connection status towards Kafka broker changes. A critical event is raised when the Kafka broker connection goes down. The event is cleared when the connection is restored.

Severity

- Critical

Auto Clearing

Yes

Clear Message

Connection to Kafka broker \$1 is up

Trigger Message

Connection to Kafka broker **{Kafka Broker IP And Port}** is down

where:

- Kafka Broker IP And Port is the Configured Kafka broker endpoint in the format IP:port or FQDN:port. If port is not provided, a default port number 9092 is assumed. where:
- Kafka Broker IP And Port is the Configured Kafka broker endpoint in the format IP:port or FQDN:port. If port is not provided, a default port number 9092 is assumed.

Potential Impact

Applications will not be able to produce/consume the messages from Kafka Broker. This could have an impact based on the use cases like EDR generation, storing charging records, streaming packets etc.

Repair Action

Make sure the Kafka broker IP and port are configured correctly and the broker is reachable.

Table of Contents

- [Gtpc Agent Interface Config Event](#)
 - [Gtpc Agent Peer Restart Event](#)
 - [Gtpc Agent Peer Status Change Event](#)
 - [Gtpc Agent Trace Config Event](#)
 - [Gtpc Agent Unique ID Not Generated Event](#)
 - [GtpcAgent EtcdAgent Event](#)
 - [GtpcAgent Peer Status Probe Event](#)
 - [GtpcAgent Peer Status Probe state change event](#)
-

Gtpc Agent Interface Config Event

Version

1.0

Description

This event indicates that an interface configuration change event has occurred.

Severity

- Info
- Major
- Critical

Auto Clearing

Yes

Clear Message

Trigger Message

The Gtpc agent detected change in interface configuration. Interface name ***{Interface} {Action}*** . Details: ***{Details}***

where:

- Interface is the Interface configured at Gtpc Agent
- Action is the Action For the configuration
- Details is the Detailed configuration entry values

Potential Impact

Consumers and GTPC peers using this interface could be affected.

Repair Action

Check the interface configuration.

Gtpc Agent Peer Restart Event

Version

1.0

Description

This event indicates that peer has restarted.

Severity

- Major

Auto Clearing

Yes

Clear Message

The Gtpc agent has received message from peer and its up

Trigger Message

The Gtpc agent detected change in restart counter. Value: **{restartcounter}** , Interface : **{Interface}** , peer ip: **{peer}** , local endpoint fqdn: **{fqdn}** , local endpoint public ip: **{localip}**

where:

- Interface is the Interface configured at Gtpc Agent
- peer is the Peer IP address
- fqdn is the Local endpoint fqdn
- localip is the Local endpoint public ip
- restartcounter is the Value of restart counter

Potential Impact

All transaction to the peer may timeout

Repair Action

Check if peer is up and reachable

Gtpc Agent Peer Status Change Event

Version

1.0

Description

This event indicates that peer status is changed to up or down. If echo timeout happens event after n3 retries then peer status is marked down and when message is received from that peer then peer status is changed to up

Severity

- Major
- Info

Auto Clearing

Yes

Clear Message

The Gtpc agent has received message from peer and its up

Trigger Message

The Gtpc agent detected n3: **{n3}** times echo timeout. Interface : **{Interface}** , peer ip: **{peer}** , local endpoint fqdn: **{fqdn}** , local endpoint public ip: **{localip}**

where:

- Interface is the Interface configured at Gtpc Agent
- peer is the Peer IP address
- fqdn is the Local endpoint fqdn
- localip is the Local endpoint public ip
- n3 is the n3 value for echo message

Potential Impact

All transactions to the peer may timeout

Repair Action

Check if peer is up and reachable

Gtpc Agent Trace Config Event

Version

1.0

Description

This event indicates that an trace configuration change event has occurred.

Severity

- Info
- Major
- Critical

Auto Clearing

Yes

Clear Message

Trigger Message

The Gtpc agent detected change in trace configuration. Trace name ***{Trace}*** ***{Action}*** . Details: ***{Details}***

where:

- Trace is the Trace configured at Gtpc Agent
- Action is the Action For the configuration
- Details is the Detailed configuration entry values

Potential Impact

Tracing capability based on the change in the trace name could be affected.

Repair Action

Check the trace configuration.

Gtpc Agent Unique ID Not Generated Event

Version

1.0

Description

This event indicates that uniqueid could not be generated for this pod.

Severity

- Critical

Auto Clearing

Yes

Clear Message

Trigger Message

The Gtpc agent pod does not have a uniqueid generated

Potential Impact

Consumer initiated messages will be dropped.

Repair Action

None.

GtpcAgent EtcdAgent Event

Version

1.0

Description

This event indicates that GtpcAgent EtcdAgent has communicated with Etcd Server.

Severity

- Info
- Major
- Critical

Auto Clearing

Yes

Clear Message

Trigger Message

The Etcd Agent event = **{Event}** for the key = **{Key}** , operation = **{Operation}** , transaction id = **{TransactionId}**

where:

- Event is the Event that triggered this log
- Key is the Key for the event that triggered this log
- Operation is the Operation performed during the event that triggered this log
- TransactionId is the Transaction ID of the transaction of the event that triggered this log

Potential Impact

Peer context creation, updation, modification might be impacted, affecting the peer-init or consumer-init call flows.

Repair Action

Check the connection to etcdServer.

GtpcAgent Peer Status Probe Event

Version

1.0

Description

This event indicates that GtpcAgent PeerStatusProbe procedure event has occurred.

Severity

- Info
- Major
- Critical

Auto Clearing

Yes

Clear Message

Trigger Message

For PeerStatusProbe procedure **{Event}** event has occurred. Summary : **{Summary}**

where:

- Event is the Event that caused this log
- Summary is the Summary of the event

Potential Impact

Peer context deletion for an unreachable peer may be affected, thereby keeping the peer context forever in the etcd.

Repair Action

Delete the peer context entry from the etcd table

GtpcAgent Peer Status Probe state change event

Version

1.0

Description

This event indicates that GtpcAgent PeerStatusProbe procedure state has changed.

Severity

- Major

Auto Clearing

Yes

Clear Message

Trigger Message

The PeerStatusProbe state has changed from ***{ExistingState}*** to ***{NewState}***

where:

- ExistingState is the Existing state that is being exited
- NewState is the New state which is being entered into

Potential Impact

Peer context deletion for an unreachable peer may be affected, thereby keeping the peer context forever in the etcd.

Repair Action

Delete the peer context entry from the etcd table

Table of Contents

- [Kafka Broker Connection Status](#)
 - [PFCP Association Status](#)
-

Kafka Broker Connection Status

Version

1.0

Description

This event is generated when the connection status towards Kafka broker changes. A critical event is raised when the Kafka broker connection goes down. The event is cleared when the connection is restored.

Severity

- Critical

Auto Clearing

Yes

Clear Message

Connection to Kafka broker \$1 is up

Trigger Message

Connection to Kafka broker **{Kafka Broker IP And Port}** is down

where:

- Kafka Broker IP And Port is the Configured Kafka broker endpoint in the format IP:port or FQDN:port. If port is not provided, a default port number 9092 is assumed. where:
- Kafka Broker IP And Port is the Configured Kafka broker endpoint in the format IP:port or FQDN:port. If port is not provided, a default port number 9092 is assumed.

Potential Impact

Applications will not be able to produce/consume the messages from Kafka Broker. This could have an impact based on the use cases like EDR generation, storing charging records, streaming packets etc.

Repair Action

Make sure the Kafka broker IP and port are configured correctly and the broker is reachable.

PFCP Association Status

Version

1.0

Description

PFCP association to a remote UPF has changed state.

Severity

- Major
- Warning

Auto Clearing

Yes

Clear Message

Association to UPF \$1/\$2 has recovered

Trigger Message

Association to UPF **{Remote Node ID}** / **{Remote Node IP Address}** has gone down

where:

- Remote Node ID is the PFCP Node ID of UPF
- Remote Node IP Address is the IP Address of UPF where:
- Remote Node ID is the PFCP Node ID of UPF
- Remote Node IP Address is the IP Address of UPF

Resource Xpath

UPF Identifier

The format of resource xpath is:

```
/pfcf/upf/[ {Remote Node ID} ]/[ {Remote Node IP Address} ]
```

where:

- Remote Node ID is the PFCP Node ID of UPF
- Remote Node IP Address is the IP Address of UPF

Alert Rules

Alert Rule 1

This event will result in raising an alert for every **PFCP Association Status** event generation based on realert configuration.

Table of Contents

- [Configuration Data File Download Status](#)
- [Configuration Schema File Upload Status](#)
- [Mongo DISK Database Connection State](#)
- [Mongo RAM Database Connection State](#)

Configuration Data File Download Status

Version

1.0

Description

Configuration data file download status.

Severity

- Major
- Info

Auto Clearing

Yes

Clear Message

Configuration data file was successfully downloaded, URL:\$1.

Trigger Message

Configuration data file download failed, URL: **{File URL}** ; diagnostic [**{Diagnostic}**]

where:

- File URL is the Configuration data file URL.
- Diagnostic is the Diagnostic (if any) that may help point to root cause. where:
- File URL is the Configuration data file URL.

Potential Impact

Application will miss latest configuration updates.

Repair Action

Fix file server connection, restart affected application pods.

Configuration Schema File Upload Status

Version

1.0

Description

Configuration schema file upload status.

Severity

- Major
- Info

Auto Clearing

Yes

Clear Message

Configuration schema file was uploaded successfully, URL:\$1.

Trigger Message

Configuration schema file upload failed, URL: **{File URL}** ; diagnostic [**{Diagnostic}**]

where:

- File URL is the Configuration schema file URL.
- Diagnostic is the Diagnostic (if any) that may help point to root cause. where:
- File URL is the Configuration schema file URL.

Potential Impact

Configuration manager pod is not upgraded with the latest version of the schema service.

Repair Action

Fix file server connection, reset application pod.

Mongo DISK Database Connection State

Version

1.0

Description

Connection to the CRUD database has changed state.

Severity

- Critical

Auto Clearing

Yes

Clear Message

Connection to the CRUD database \$1/\$2:\$3 is UP.

Trigger Message

Connection to the CRUD database **{Database ConfigMap}** / **{Database IP Address}** : **{Database IP Port}** is DOWN; diagnostic [**{Diagnostic}**]

where:

- Database ConfigMap is the ConfigMap which contains configurable database connectivity details.
 - Database IP Address is the Configured IP address, from the ConfigMap, used to connect to the database.
 - Database IP Port is the Configured IP port, from the ConfigMap, used to connect to the database.
 - Diagnostic is the Diagnostic (if any) that may help point to root cause. where:
 - Database ConfigMap is the ConfigMap which contains configurable database connectivity details.
 - Database IP Address is the Configured IP address, from the ConfigMap, used to connect to the database.
 - Database IP Port is the Configured IP port, from the ConfigMap, used to connect to the database.
-

Mongo RAM Database Connection State

Version

1.0

Description

Connection to the CRUD database has changed state.

Severity

- Critical

Auto Clearing

Yes

Clear Message

Connection to the CRUD database \$1/\$2:\$3 is UP.

Trigger Message

Connection to the CRUD database **{Database ConfigMap} / {Database IP Address} : {Database IP Port}** is DOWN; diagnostic [**{Diagnostic}**]

where:

- Database ConfigMap is the ConfigMap which contains configurable database connectivity details.
- Database IP Address is the Configured IP address, from the ConfigMap, used to connect to the database.
- Database IP Port is the Configured IP port, from the ConfigMap, used to connect to the database.
- Diagnostic is the Diagnostic (if any) that may help point to root cause. where:
- Database ConfigMap is the ConfigMap which contains configurable database connectivity details.

- Database IP Address is the Configured IP address, from the ConfigMap, used to connect to the database.
- Database IP Port is the Configured IP port, from the ConfigMap, used to connect to the database.

Table of Contents

- [INTFMGR_CFGMGR_REGISTRATION](#)
 - [INTFMGR_CONFIG_VALIDATION](#)
 - [INTFMGR_DATAPLANE_SERVICE](#)
 - [INTFMGR_ETCD_SERVICE](#)
 - [INTFMGR_RESOURCE_EXHAUSTION](#)
 - [INTFMGR_SERVICE_START](#)
 - [INTFMGR_SERVREG_PUBLISH](#)
 - [INTFMGR_SERVREG_SUBSCRIBE](#)
-

INTFMGR_CFGMGR_REGISTRATION

Version

1.0

Description

Reports Interface manager's progress in registering with the Configuration Manager.

Severity

- Critical
- Major
- Minor

Status

Current

Auto Clearing

Yes

Clear Message

Registered with Configuration Manager via its sidecar. URL of request: \$1. Request body: \$2.

Root Cause

Communication problem, or Configuration Manager is unavailable.

Trigger Message

Failed to register with Configuration Manager via its sidecar. URL of request: ***{requestUrl}*** . Request body: ***{requestObject}*** . ***{error}***

where:

- requestUrl is the URL of ConfigMgr sidecar.
- requestObject is the JSON body of registration request.
- error is the Explanation of registration error. where:
 - requestUrl is the URL of ConfigMgr sidecar.
 - requestObject is the JSON body of registration request.

Potential Impact

If Interface Manager fails to register with Config Manager, no network configuration will be deployed in the UPF. The UPF will be unable to process traffic.

Repair Action

Verify Configuration Manager status, and its reachability.

Resource Xpath

URL used to reach the Configuration Manager sidecar.

The format of resource xpath is:

```
{requestUrl}
```

where:

- requestUrl is the URL of ConfigMgr sidecar.

INTFMGR_CONFIG_VALIDATION

Version

1.0

Description

Reports inconsistencies detected in the Interface Manager configuration.

Severity

- Major

Status

Current

Auto Clearing

Yes

Clear Message

Configuration object \$1 passed validation checks.

Root Cause

Configuration error.

Trigger Message

Configuration object **{xpath}** failed validation check(s). **{error}**

where:

- xpath is the Xpath to problematic configuration item.
- error is the Explanation of configuration problem. where:
- xpath is the Xpath to problematic configuration item.

Potential Impact

If configuratin inconsistencies are found there may be insufficient networking resources available to support all Dataplane Services. UPF traffic forwarding may be impacted.

Repair Action

Correct the configuration.

Resource Xpath

Path to an attribute in the Interface Manager configuration.

The format of resource xpath is:

{xpath}

where:

- xpath is the Xpath to problematic configuration item.

INTFMGR_DATAPLANE_SERVICE

Version

1.0

Description

Reports the availability of Dataplane services and their external interfaces.

Severity

- Critical

Status

Current

Auto Clearing

Yes

Clear Message

External interface(s) detected. Number of Dataplane Services: \$1. Number of external interfaces: \$2.

Root Cause

Communication with Service Registry, or no Dataplane Agents in UPF, or Dataplane Agents failed to publish to Service Registry, or Dataplane Agents have no external interfaces.

Trigger Message

No external interfaces detected. Number of Dataplane Services: **{dpaCount}** . Number of external interfaces: **{interfaceCount}** .

where:

- dpaCount is the Number of Dataplane Agents detected.
- interfaceCount is the Total count of external interfaces available on all Dataplane Agents. where:
- dpaCount is the Number of Dataplane Agents detected.
- interfaceCount is the Total count of external interfaces available on all Dataplane Agents.

Potential Impact

If no external interfaces are found, UPF will be unable to process traffic.

Repair Action

Verify Dataplane Agent existence; Verify external interfaces are known to Dataplane Agent(s). Verify Service Registry status, and its reachability from Interface Manager and Dataplane Agents(s).

INTFMGR_ETCD_SERVICE

Version

1.0

Description

Reports the availability of the etcd service.

Severity

- Critical

Status

Current

Auto Clearing

Yes

Clear Message

Etcd Service is available. \$1

Root Cause

Communication with etcd Server, or no etcd Server failure.

Trigger Message

Etcd Service is unavailable. ***{optionalMsg}***

where:

- optionalMsg is the Optional message. Normally not used. where:
- optionalMsg is the Optional message. Normally not used.

Potential Impact

If the etcd service is unavailable, no networking configuration will be deployed in the UPF. The UPF will be unable to process traffic.

Repair Action

Verify etcd Server availability, and its reachability from Interface Manager.

INTFMGR_RESOURCE_EXHAUSTION

Version

1.0

Description

Reports the lack of a configuration resource required to correctly deploy a VLAN sub-interface on a Dataplane Service.

Severity

- Major

Status

Current

Auto Clearing

Yes

Clear Message

Configuration object \$1 has sufficient resources to deploy a VLAN sub-interface on interface \$2 of Dataplane Service \$3.

Root Cause

Insufficient configuration resources.

Trigger Message

Configuration object **{xpath}** has insufficient resources to deploy a VLAN sub-interface on interface **{interfaceName}** of Dataplane Service **{podServiceId}** .

where:

- xpath is the Xpath to problematic configuration item.
- interfaceName is the Nama of interface that hosts the VLAN sub-interface.
- podServiceId is the ServiceId of Dataplane Service affected by the resource shortage. where:
- xpath is the Xpath to problematic configuration item.
- interfaceName is the Nama of interface that hosts the VLAN sub-interface.
- podServiceId is the ServiceId of Dataplane Service affected by the resource shortage.

Potential Impact

A Dataplane Service will not be correctly configured. Some traffic forwarding will be impacted.

Repair Action

Correct the configuration. The VLAN address template should be adjusted to provide more addresses.

Resource Xpath

Path to an attribute in the Interface Manager configuration.

The format of resource xpath is:

```
{xpath}
```

where:

- xpath is the Xpath to problematic configuration item.

INTFMGR_SERVICE_START

Version

1.0

Description

Reports problems with environment required to correctly initialize the Interface Manager service.

Severity

- Critical

Status

Current

Auto Clearing

Yes

Clear Message

Interface Manager service in pod: \$1, container: \$2, initialized correctly.

Root Cause

Missing or invalid environment variables in the Interface Manager container.

Trigger Message

Interface Manager service in pod: **{podName}** , container: **{containerName}** , did not initialize correctly.
{variables}

where:

- podName is the Name of Pod containing Interface Manager.
- containerName is the Name of Interface Manager container.
- variables is the List of problematic environment variables. where:
- podName is the Name of Pod containing Interface Manager.
- containerName is the Name of Interface Manager container.

Potential Impact

Interface Manager will not operate correctly. Some, or all, functionality will be impacted.

Repair Action

Correct the environment variables.

INTFMGR_SERVREG_PUBLISH

Version

1.0

Description

Reports Interface manager's progress in publishing its service to the Service Registry.

Severity

- Critical
- Major
- Minor

Status

Current

Auto Clearing

Yes

Clear Message

Published Interface Manager \$4 status to Service Registry, via its sidecar. URL of request: \$1. Request body: \$2.

Root Cause

Communication problem, or Service Registry is unavailable.

Trigger Message

Failed to publish Interface Manager **{status}** status to Service Registry, via its sidecar. URL of request: **{requestUrl}** . Request body: **{requestObject}** . **{error}**

where:

- requestUrl is the URL of Service Registry sidecar.
- requestObject is the JSON body of publish request.

- error is the Explanation of publish error.
- status is the Status of service. PRESENT indicates service is starting. READY indicates service is available for use. where:
- requestUrl is the URL of Service Registry sidecar.
- requestObject is the JSON body of publish request.
- status is the Status of service. PRESENT indicates service is starting. READY indicates service is available for use.

Potential Impact

If Interface Manager fails to publish to the Service Registry, other services dependent on Interface Manager may not function correctly. UPF may be partially functional.

Repair Action

Verify Service Registry status, and its reachability.

Resource Xpath

URL used to reach the Service Registry sidecar.

The format of resource xpath is:

```
{requestUrl}
```

where:

- requestUrl is the URL of Service Registry sidecar.

INTFMGR_SERVREG_SUBSCRIBE

Version

1.0

Description

Reports Interface manager's progress in subscribing to the Service Registry to learn of Dataplane Services.

Severity

- Critical
- Major
- Minor

Status

Current

Auto Clearing

Yes

Clear Message

Subscribed to Service Registry, via its sidecar, for Dataplane Agent service notifications. URL of request: \$1.
Request body: \$2.

Root Cause

Communication problem, or Service Registry is unavailable.

Trigger Message

Failed to subscribe to Service Registry, via its sidecar, for Dataplane Agent service notifications. URL of request: **`{requestUrl}`** . Request body: **`{requestObject}`** . **`{error}`**

where:

- requestUrl is the URL of Service Registry sidecar.
- requestObject is the JSON body of subscription request.
- error is the Explanation of subscription error. where:
- requestUrl is the URL of Service Registry sidecar.
- requestObject is the JSON body of subscription request.

Potential Impact

If Interface Manager fails to subscribe to the Service Registry, no networking configuration will be deployed in the UPF. The UPF will be unable to process traffic.

Repair Action

Verify Service Registry status, and its reachability.

Resource Xpath

URL used to reach the Service Registry sidecar.

The format of resource xpath is:

```
{requestUrl}
```

where:

- requestUrl is the URL of Service Registry sidecar.

Table of Contents

- [Istio Gateway Downstream Request HTTP Protocol Error Event](#)

- [Istio Gateway No healthy upstream host Event](#)
- [Istio Gateway Route not found Event](#)

Istio Gateway Downstream Request HTTP Protocol Error Event

Version

1.0

Description

This event alert indicates that the istio ingress gateway has received request with HTTP URL that is unknown to the system

Severity

- Warning

Auto Clearing

No

Root Cause

This could be due to a client request that does not conform to the HTTP protocol standards. It could be due to a malformed request, missing or incorrect headers, invalid characters in the URL or request body, etc.

Trigger Message

The downstream request had an HTTP protocol error

Potential Impact

The impact may vary depending on the nature of the request and the criticality of the resource requested. The client may need to modify the request to conform to the HTTP protocol standards and resend the request to obtain the desired response.

Repair Action

Check the syntax and format of the HTTP request being sent by the client to ensure that it conforms to the HTTP protocol standards.

Alert Rules

Alert Rule 1

This event will result in raising an alert for every ***Istio Gateway Downstream Request HTTP Protocol Error Event*** event generation based on realert configuration.

Istio Gateway No healthy upstream host Event

Version

1.0

Description

This event alert indicates that there were no healthy upstream hosts in the upstream cluster.

Severity

- Warning

Auto Clearing

No

Root Cause

Few of the possible reasons are Incorrect hostname in the request, DNS configuration issues, Unavailable upstream services, Network connectivity issues, or Incorrect Istio configuration

Trigger Message

GET ***{upstream host}*** 503 No healthy upstream hosts in upstream cluster

where:

- upstream host is the Route that egress gateway is trying to reach. It is the URL of the request being tried to reach.

Potential Impact

Unavailable services, Increased latency, and degraded performance of downstream services.

Repair Action

Verify the hostname used in the request, Check DNS configuration, Check the availability and health of the upstream services, Check for network issues, or Restart Istio components.

Alert Rules

Alert Rule 1

This event will result in raising an alert for every ***Istio Gateway No healthy upstream host Event*** event generation based on realert configuration.

Istio Gateway Route not found Event

Version

1.0

Description

This event alert indicates that the istio gateway has received a request with the destination path that is unknown to the system

Severity

- Warning

Auto Clearing

No

Root Cause

This can occur for a variety of reasons, such as, Istio gateway is misconfigured, requested URL is invalid, requested URL does not match any of the available routes, or when the requested service is not available

Trigger Message

POST ***{Desination route}*** 404 NR route_not_found

where:

- Desination route is the Route that istio gateway is trying to reach.

Potential Impact

This error can cause user-facing applications to fail or malfunction, leading to a poor user experience. Additionally, it could also cause issues with downstream services that depend on the missing resource or endpoint.

Repair Action

Few of the possible repair actions are: Check that the requested resource or endpoint actually exists and is reachable within the service mesh, Check routing rules and virtual services, Check the service registry and discovery

Alert Rules

Alert Rule 1

This event will result in raising an alert for every **Istio Gateway Route not found Event** event generation based on realert configuration.

Table of Contents

- [Kafka Broker Connection Status](#)
-

Kafka Broker Connection Status

Version

1.0

Description

This event is generated when the connection status towards Kafka broker changes. A critical event is raised when the Kafka broker connection goes down. The event is cleared when the connection is restored.

Severity

- Critical

Auto Clearing

Yes

Clear Message

Connection to Kafka broker \$1 is up

Trigger Message

Connection to Kafka broker **{Kafka Broker IP And Port}** is down

where:

- Kafka Broker IP And Port is the Configured Kafka broker endpoint in the format IP:port or FQDN:port. If port is not provided, a default port number 9092 is assumed. where:
- Kafka Broker IP And Port is the Configured Kafka broker endpoint in the format IP:port or FQDN:port. If port is not provided, a default port number 9092 is assumed.

Potential Impact

Applications will not be able to produce/consume the messages from Kafka Broker. This could have an impact based on the use cases like EDR generation, storing charging records, streaming packets etc.

Repair Action

Make sure the Kafka broker IP and port are configured correctly and the broker is reachable.

Table of Contents

- [Kafka Broker Connection Status](#)
 - [Kafka Replay Agent Configuration Event](#)
-

Kafka Broker Connection Status

Version

1.0

Description

This event is generated when the connection status towards Kafka broker changes. A critical event is raised when the Kafka broker connection goes down. The event is cleared when the connection is restored.

Severity

- Critical

Auto Clearing

Yes

Clear Message

Connection to Kafka broker \$1 is up

Trigger Message

Connection to Kafka broker **{Kafka Broker IP And Port}** is down

where:

- Kafka Broker IP And Port is the Configured Kafka broker endpoint in the format IP:port or FQDN:port. If port is not provided, a default port number 9092 is assumed. where:
- Kafka Broker IP And Port is the Configured Kafka broker endpoint in the format IP:port or FQDN:port. If port is not provided, a default port number 9092 is assumed.

Potential Impact

Applications will not be able to produce/consume the messages from Kafka Broker. This could have an impact based on the use cases like EDR generation, storing charging records, streaming packets etc.

Repair Action

Make sure the Kafka broker IP and port are configured correctly and the broker is reachable.

Kafka Replay Agent Configuration Event

Version

1.0

Description

This event indicates the provisioning of replay configuration for the Kafka Replay Agent to replay the messages to the destination based on the received configuration

Severity

- Info

Auto Clearing

No

Trigger Message

Replay Configuration- Event Type: ***{Replay Configuration Event Type}*** , Topic Name: ***{Topic Name}*** , Configuration: ***{Replay Configuration}***

where:

- Replay Configuration Event Type is the Type of Replay Configuration Event: CREATE/UPDATE/DELETE
- Topic Name is the Name of the topic name for which the event is received
- Replay Configuration is the The replay configuration parameters recieved for the given topic

Table of Contents

- [UPF Ip Pool Config Error Event](#)
-

UPF Ip Pool Config Error Event

Version

1.0

Description

This event indicates that the given UPF received wrong Ip Pool Configuration and it won't be applied

Severity

- Major

Auto Clearing

Yes

Clear Message

Correct configurations received for the Ip pool

Trigger Message

{Error Message}

- Error Message is the Error Message for Wrong Config

Potential Impact

UPF will not be able to process the new config

Repair Action

Check and correct the ip-pool.

Resource Xpath

Ip Pool in Network Context for which config is wrong.

The format of resource xpath is:

```
/upf/service-construct/network-context/ {Network Context Name} /ip-pool/ {Pool name}
```

where:

- Network Context Name is the Network-Context Name
- Pool name is the IP-Pool entry with pool name.

Table of Contents

- [Kafka Broker Connection Status](#)
 - [Possible Dead Lock Detection Event](#)
 - [Unexpected Container Restart Detection Event](#)
-

Kafka Broker Connection Status

Version

1.0

Description

This event is generated when the connection status towards Kafka broker changes. A critical event is raised when the Kafka broker connection goes down. The event is cleared when the connection is restored.

Severity

- Critical

Auto Clearing

Yes

Clear Message

Connection to Kafka broker \$1 is up

Trigger Message

Connection to Kafka broker **{Kafka Broker IP And Port}** is down

where:

- Kafka Broker IP And Port is the Configured Kafka broker endpoint in the format IP:port or FQDN:port. If port is not provided, a default port number 9092 is assumed. where:
- Kafka Broker IP And Port is the Configured Kafka broker endpoint in the format IP:port or FQDN:port. If port is not provided, a default port number 9092 is assumed.

Potential Impact

Applications will not be able to produce/consume the messages from Kafka Broker. This could have an impact based on the use cases like EDR generation, storing charging records, streaming packets etc.

Repair Action

Make sure the Kafka broker IP and port are configured correctly and the broker is reachable.

Alert Rules

Alert Rule 1

This event will result in raising an alert due to change with respect to **currentState** field based on realert configuration. The rising of this alert is for each unique value of **kubernetes.namespace_name** field with realert time counted separately.

Possible Dead Lock Detection Event

Version

1.0

Description

This event indicates that an unresponsive thread detected probably due to a dead-lock in an application process.

Severity

- Major

Auto Clearing

No

Trigger Message

Possible dead-lock detected in process ***{Process Name}***

where:

- Process Name is the Name of the process

Potential Impact

This results in application pod restarts which could in turn result in service outage as the application could become unavailable to provide its service.

Repair Action

Identify the root cause of the dead-lock in the application using the generated coredumps to fix it

Alert Rules

Alert Rule 1

This event will result in raising an alert for every ***Possible Dead Lock Detection Event*** event generation based on realert configuration. The rising of this alert is for each unique value of ***processName***, *** with realert time counted separately.

Unexpected Container Restart Detection Event

Version

1.0

Description

This event indicates that an unexpected container restart is detected resulting in the pod termination. Possible reasons for unexpected container restart includes the given container hitting OOM error.

Severity

- Major

Auto Clearing

No

Trigger Message

Container [**{Container Name}**] got restarted unexpectedly

where:

- Container Name is the Name of the container which restarted.

Potential Impact

Application container restarts could result in service impact as they could be unavailable to provide its service.

Repair Action

Identify the application name and the root cause of the container restart to fix the issue.

Alert Rules

Alert Rule 1

This event will result in raising an alert for every **Unexpected Container Restart Detection Event** event generation based on realert configuration. The rising of this alert is for each unique value of **containerName**, *** with realert time counted separately.

Table of Contents

- [eNB Reachability](#)
-

eNB Reachability

Version

1.0

Description

Transport Network Layer (TNL) association status to an eNB has changed. This event is set approximately 10 seconds after the SCTP connection to the eNB goes down. The alarm information indicates the remote IP and port of the eNB.

It is automatically cleared when the eNB is up. Manually clear the alarm if the eNB is not expected to return to service or is removed from the network.

Severity

- Major

Auto Clearing

Yes

Clear Message

Connectivity is up to eNB with Global eNB ID: \$1, eNB Name: \$2; TNL association with IP address/addresses \$3 and port \$4 has connected.

Trigger Message

Connectivity is down to eNB with Global eNB ID: **{Global eNB ID}**, eNB Name: **{eNB Name}**; TNL association with IP address/addresses **{TNL association IP Address}** and port **{TNL association IP Port}** has gone down.

where:

- Global eNB ID is the eNB identifier in the form of a Global eNB ID. This is comprised of PLMN, a unique eNB instance ID and the eNB type.
- eNB Name is the Name of the eNB, or 'not-available' if not provided by the eNB.
- TNL association IP Address is the IP of the eNB.
- TNL association IP Port is the Port of the eNB. where:
- Global eNB ID is the eNB identifier in the form of a Global eNB ID. This is comprised of PLMN, a unique eNB instance ID and the eNB type.
- eNB Name is the Name of the eNB, or 'not-available' if not provided by the eNB.
- TNL association IP Address is the IP of the eNB.
- TNL association IP Port is the Port of the eNB.

Potential Impact

Loss of service in the area covered by this eNB.

Repair Action

Verify the health of the eNB and IP network.

Resource Xpath

Global eNB ID and eNB Name

The format of resource xpath is:

```
/mme/enb/[ {Global eNB ID} ]/[ {eNB Name} ]
```

where:

- Global eNB ID is the eNB identifier in the form of a Global eNB ID. This is comprised of PLMN, a unique eNB instance ID and the eNB type.
- eNB Name is the Name of the eNB, or 'not-available' if not provided by the eNB.

Alert Rules

Alert Rule 1

This event will result in raising an alert for every **eNB Reachability** event generation based on realert configuration. The rising of this alert is for each unique value of **severity** and **globalEnbId** fields with realert time counted separately.

Table of Contents

- [Kafka Broker Connection Status](#)

Kafka Broker Connection Status

Version

1.0

Description

This event is generated when the connection status towards Kafka broker changes. A critical event is raised when the Kafka broker connection goes down. The event is cleared when the connection is restored.

Severity

- Critical

Auto Clearing

Yes

Clear Message

Connection to Kafka broker \$1 is up

Trigger Message

Connection to Kafka broker {Kafka Broker IP And Port} is down

where:

- Kafka Broker IP And Port is the Configured Kafka broker endpoint in the format IP:port or FQDN:port. If port is not provided, a default port number 9092 is assumed. where:
- Kafka Broker IP And Port is the Configured Kafka broker endpoint in the format IP:port or FQDN:port. If port is not provided, a default port number 9092 is assumed.

Potential Impact

Applications will not be able to produce/consume the messages from Kafka Broker. This could have an impact based on the use cases like EDR generation, storing charging records, streaming packets etc.

Repair Action

Make sure the Kafka broker IP and port are configured correctly and the broker is reachable.

Table of Contents

- [NF availability status](#)
 - [Peer NRF Down Event](#)
-

NF availability status

Version

1.0

Description

This event indicates that an NF is unavailable due to its status change to either suspended or undiscoverable state.

Severity

- Major

Auto Clearing

Yes

Clear Message

NF status has changed to registered or deregistered state. Affected NF Instance ID: \$1; NF type: \$2; New NF status: \$3

Trigger Message

NF is unavailable due to its NF status has changed to suspended or undiscoverable state. Affected NF Instance ID: **{NF instance ID}** ; NF type: **{NF type}** ; New NF status: **{New NF status}**

where:

- NF instance ID is the The NF instance identifier in the form of UUID
- NF type is the The NF type such as AMF, SMF or UPF etc.
- New NF status is the The new status of the NF where:
- NF instance ID is the The NF instance identifier in the form of UUID
- NF type is the The NF type such as AMF, SMF or UPF etc.
- New NF status is the The new status of the NF

Potential Impact

Loss of services provided by the NF

Repair Action

Verify the health and status of the NF. The NF unavailable event is cleared when the NF status has changed to registered or deregistered state.

Resource Xpath

NF Instance Identifier

The format of resource xpath is:

```
/nrf/nf/[ {NF instance ID} ]
```

where:

- NF instance ID is the The NF instance identifier in the form of UUID

Alert Rules

Alert Rule 1

This event will result in raising an alert for every **NF availability status** event generation based on realert configuration. The rising of this alert is for each unique value of **severity** and **nfnstancelid** fields with realert time counted separately.

Peer NRF Down Event

Version

1.0

Description

This event indicates that one of the peer NRF status has changed to DOWN. The peer status is set to DOWN on any failure response of 4xx and 5xx in the NRF peer heartbeat response other than 460 and 404. The alarm information includes the peer NRF IP address or FQDN, and the port number.

It is automatically cleared when the peer is returned back to service, or removed from the peer-set configuration.

This alarm is only applicable if the NRF Geo-redundancy feature is enabled by providing the nrf-set data in the nf-inv-mgr configuration

Severity

- Major

Auto Clearing

Yes

Clear Message

Peer NRF is UP. Peer instance id: \$1; IP or FQDN: \$2 and port: \$3.

Trigger Message

Peer NRF is DOWN. Affected peer instance id: **{NRF instance ID}** ; IP or FQDN: **{IP address or FQDN}** and port: **{Port}** .

where:

- NRF instance ID is the The NRF instance identifier in the form of UUID.
- IP address or FQDN is the The IP address or FQDN of the peer NRF.
- Port is the The port number of the peer NRF. where:
- NRF instance ID is the The NRF instance identifier in the form of UUID.
- IP address or FQDN is the The IP address or FQDN of the peer NRF.
- Port is the The port number of the peer NRF.

Potential Impact

Loss of services provided by this peer NRF.

Repair Action

Verify the health of the peer NRF and IP network. The NRF redundancy will automatically recover sync when it is recovered and back in service

Resource Xpath

Peer NRF Instance Identifier

The format of resource xpath is:

```
/nrf/peer/[ {NRF instance ID} ]
```

where:

- NRF instance ID is the The NRF instance identifier in the form of UUID.

Alert Rules

Alert Rule 1

This event will result in raising an alert for every **Peer NRF Down Event** event generation based on realert configuration. The rising of this alert is for each unique value of **severity** and **nrfInstanceld** fields with realert time counted separately.

Table of Contents

- [Kafka Broker Connection Status](#)
 - [NRF Down Event](#)
-

Kafka Broker Connection Status

Version

1.0

Description

This event is generated when the connection status towards Kafka broker changes. A critical event is raised when the Kafka broker connection goes down. The event is cleared when the connection is restored.

Severity

- Critical

Auto Clearing

Yes

Clear Message

Connection to Kafka broker \$1 is up

Trigger Message

Connection to Kafka broker {**Kafka Broker IP And Port**} is down

where:

- Kafka Broker IP And Port is the Configured Kafka broker endpoint in the format IP:port or FQDN:port. If port is not provided, a default port number 9092 is assumed. where:
- Kafka Broker IP And Port is the Configured Kafka broker endpoint in the format IP:port or FQDN:port. If port is not provided, a default port number 9092 is assumed.

Potential Impact

Applications will not be able to produce/consume the messages from Kafka Broker. This could have an impact based on the use cases like EDR generation, storing charging records, streaming packets etc.

Repair Action

Make sure the Kafka broker IP and port are configured correctly and the broker is reachable.

NRF Down Event

Version

1.0

Description

This event indicates that one of the NRF status has changed to DOWN. The status is set to DOWN on any failure response configured as part of the nrf-profile configuration. The alarm information includes the peer NRF IP address or FQDN, and the port number.

It is automatically cleared when the peer is returned back to service, or removed from the nrf-profile.

This alarm is only applicable if the NRF-Agent is configured in 'primary-secondary' mode under nrf-agent nrf-redundancy-mode

Severity

- Major

Auto Clearing

Yes

Clear Message

NRF is UP. Instance id: \$1; ServiceName: \$2.

Trigger Message

NRF is DOWN. Affected NRF instance id: **{NRF instance ID}** ; ServiceName: **{NRF service name}**

where:

- NRF instance ID is the The NRF instance identifier in the form of UUID.
- NRF service name is the The NRF service name. where:
- NRF instance ID is the The NRF instance identifier in the form of UUID.
- NRF service name is the The NRF service name.

Potential Impact

Loss of services provided by this peer NRF.

Repair Action

Verify the health of the peer NRF and IP network. The keep alive towards the configred NRF will continue

Resource Xpath

NRF Instance Identifier

The format of resource xpath is:

```
/nrf/peer/[ {NRF instance ID} ]
```

where:

- NRF instance ID is the The NRF instance identifier in the form of UUID.

Alert Rules

Alert Rule 1

This event will result in raising an alert for every **NRF Down Event** event generation based on realert configuration. The rising of this alert is for each unique value of **severity** and **nrfInstanceld** fields with realert time counted separately.

Table of Contents

- [Consul Read Status](#)
 - [Critical Service not Available](#)
 - [NF Management Status](#)
 - [Unstable NF due to Frequent Restart](#)
 - [Unstable NF due to Readiness Failure](#)
-

Consul Read Status

Version

1.0

Description

Reading Previously Stored Status from Consul.

Severity

- Info

Auto Clearing

Yes

Clear Message

Consul Read Response:\$1 at:\$2 NF:\$4 is using the :\$3 InstanceID

Trigger Message

Consul Read Response: **{Consul Response}** at: **{Event TimeStamp}** NF: **{Network Function Name}** is using the : **{Info Message}** InstanceID

where:

- Consul Response is the Consul Read HTTP response
- Event TimeStamp is the Event Timestamp
- Info Message is the Info about the InstanceID (static/dynamic/existing)
- Network Function Name is the Network Function Name where:
- Consul Response is the Consul Read HTTP response
- Event TimeStamp is the Event Timestamp
- Info Message is the Info about the InstanceID (static/dynamic/existing)
- Network Function Name is the Network Function Name

Repair Action

Informative Event about the Consul Read Status, No Action Required

Alert Rules

Alert Rule 1

This event will result in raising an alert for every **Consul Read Status** event generation based on realert configuration. The rising of this alert is for each unique value of **severity**, **podName** and **networkFunctionName** fields with realert time counted separately.

Critical Service not Available

Version

1.0

Description

Critical Service is Unavailable

Severity

- Critical

Auto Clearing

Yes

Clear Message

Critical Service :\$1 has become Available (\$4) at :\$2 NF:\$3

Trigger Message

Critical Service : **{Critical Service Name}** is unavailable (**{Service Status}**) at : **{Event TimeStamp}** NF: **{Network Function Name}**

where:

- Critical Service Name is the Critical Service Name
- Event TimeStamp is the Event Timestamp
- Network Function Name is the Network Function Name
- Service Status is the Available/Required where:
- Critical Service Name is the Critical Service Name
- Event TimeStamp is the Event Timestamp
- Network Function Name is the Network Function Name
- Service Status is the Available/Required

Repair Action

Check if all the required critical services are available

Alert Rules

Alert Rule 1

This event will result in raising an alert for every **Critical Service not Available** event generation based on realert configuration. The rising of this alert is for each unique value of **severity**, **criticalServiceName** and **networkFunctionName** fields with realert time counted separately.

NF Management Status

Version

1.0

Description

NF Status towards NRF

Severity

- Info

Auto Clearing

Yes

Clear Message

NF:\$4 has updated NRF with status:\$1 at:\$2 InstanceID:\$3

Trigger Message

NF: **{Network Function Name}** has updated NRF with status: **{NfMgmt Status}** at: **{Event TimeStamp}**
InstanceID: **{NF Instance ID}**

where:

- NfMgmt Status is the Nf Management Interface Status(Registered/DeRegistered/Maintenance)
- Event TimeStamp is the Event Timestamp
- NF Instance ID is the Network Function Instance ID
- Network Function Name is the Network Function Name where:
- NfMgmt Status is the Nf Management Interface Status(Registered/DeRegistered/Maintenance)
- Event TimeStamp is the Event Timestamp
- NF Instance ID is the Network Function Instance ID
- Network Function Name is the Network Function Name

Repair Action

Informative Event about the NF Status, No Action Required

Alert Rules

Alert Rule 1

This event will result in raising an alert for every **NF Management Status** event generation based on realert configuration. The rising of this alert is for each unique value of **networkFunctionName**, *** with realert time counted separately.

Unstable NF due to Frequent Restart

Version

1.0

Description

Network Function is unstable due to Frequent Restarts

Severity

- Critical

Auto Clearing

Yes

Clear Message

Critical Service:\$1 is stable.

Trigger Message

Due to Critical Service: **{Critical Service Name}** , NF: **{Network Function Name}** is unstable, timestamp: **{Event TimeStamp}**

where:

- Critical Service Name is the Critical Service Name
- Event TimeStamp is the Event Timestamp
- Network Function Name is the Network Function Name where:
- Critical Service Name is the Critical Service Name

Repair Action

Check why Critical Service instances are restarting

Alert Rules

Alert Rule 1

This event will result in raising an alert for every **Unstable NF due to Frequent Restart** event generation based on realert configuration. The rising of this alert is for each unique value of **severity**, **criticalServiceName** and **networkFunctionName** fields with realert time counted separately.

Unstable NF due to Readiness Failure

Version

1.0

Description

Network Function is unstable due to Readiness Failure

Severity

- Critical

Auto Clearing

Yes

Clear Message

Critical Service:\$1 is stable.

Trigger Message

Due to Critical Service: **{Critical Service Name}** , NF: **{Network Function Name}** is unstable, timestamp: **{Event TimeStamp}**

where:

- Critical Service Name is the Critical Service Name
- Event TimeStamp is the Event Timestamp
- Network Function Name is the Network Function Name where:
- Critical Service Name is the Critical Service Name

Repair Action

Check why there are Readiness Probe failures for the Critical Service

Alert Rules

Alert Rule 1

This event will result in raising an alert for every **Unstable NF due to Readiness Failure** event generation based on realert configuration. The rising of this alert is for each unique value of **severity**, **criticalServiceName** and **networkFunctionName** fields with realert time counted separately.

Table of Contents

- [Consul Read Status](#)
 - [Critical Service not Available](#)
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 - [Unstable NF due to Readiness Failure](#)
-

Consul Read Status

Version

1.0

Description

Reading Previously Stored Status from Consul.

Severity

- Info

Auto Clearing

Yes

Clear Message

Consul Read Response:\$1 at:\$2 NF:\$4 is using the :\$3 InstanceID

Trigger Message

Consul Read Response: **{Consul Response}** at: **{Event TimeStamp}** NF: **{Network Function Name}** is using the : **{Info Message}** InstanceID

where:

- Consul Response is the Consul Read HTTP response
- Event TimeStamp is the Event Timestamp
- Info Message is the Info about the InstanceID (static/dynamic/existing)
- Network Function Name is the Network Function Name where:
- Consul Response is the Consul Read HTTP response
- Event TimeStamp is the Event Timestamp
- Info Message is the Info about the InstanceID (static/dynamic/existing)
- Network Function Name is the Network Function Name

Repair Action

Informative Event about the Consul Read Status, No Action Required

Alert Rules

Alert Rule 1

This event will result in raising an alert for every **Consul Read Status** event generation based on realert configuration. The rising of this alert is for each unique value of **severity**, **podName** and **networkFunctionName** fields with realert time counted separately.

Critical Service not Available

Version

1.0

Description

Critical Service is Unavailable

Severity

- Critical

Auto Clearing

Yes

Clear Message

Critical Service :\$1 has become Available (\$4) at :\$2 NF:\$3

Trigger Message

Critical Service : **{Critical Service Name}** is unavailable (**{Service Status}**) at : **{Event TimeStamp}** NF: **{Network Function Name}**

where:

- Critical Service Name is the Critical Service Name
- Event TimeStamp is the Event Timestamp
- Network Function Name is the Network Function Name
- Service Status is the Available/Required where:
 - Critical Service Name is the Critical Service Name
 - Event TimeStamp is the Event Timestamp
 - Network Function Name is the Network Function Name
 - Service Status is the Available/Required

Repair Action

Check if all the required critical services are available

Alert Rules

Alert Rule 1

This event will result in raising an alert for every **Critical Service not Available** event generation based on realert configuration. The rising of this alert is for each unique value of **severity**, **criticalServiceName** and **networkFunctionName** fields with realert time counted separately.

NF Management Status

Version

1.0

Description

NF Status towards NRF

Severity

- Info

Auto Clearing

Yes

Clear Message

NF:\$4 has updated NRF with status:\$1 at:\$2 InstanceID:\$3

Trigger Message

NF: **{Network Function Name}** has updated NRF with status: **{NfMgmt Status}** at: **{Event TimeStamp}**
InstanceID: **{NF Instance ID}**

where:

- NfMgmt Status is the Nf Management Interface Status(Registered/DeRegistered/Maintenance)
- Event TimeStamp is the Event Timestamp
- NF Instance ID is the Network Function Instance ID
- Network Function Name is the Network Function Name where:
- NfMgmt Status is the Nf Management Interface Status(Registered/DeRegistered/Maintenance)
- Event TimeStamp is the Event Timestamp
- NF Instance ID is the Network Function Instance ID
- Network Function Name is the Network Function Name

Repair Action

Informative Event about the NF Status, No Action Required

Alert Rules

Alert Rule 1

This event will result in raising an alert for every **NF Management Status** event generation based on realert configuration. The rising of this alert is for each unique value of **networkFunctionName**, *** with realert time counted separately.

Unstable NF due to Frequent Restart

Version

1.0

Description

Network Function is unstable due to Frequent Restarts

Severity

- Critical

Auto Clearing

Yes

Clear Message

Critical Service:\$1 is stable.

Trigger Message

Due to Critical Service: **{Critical Service Name}** , NF: **{Network Function Name}** is unstable, timestamp: **{Event TimeStamp}**

where:

- Critical Service Name is the Critical Service Name
- Event TimeStamp is the Event Timestamp
- Network Function Name is the Network Function Name where:
- Critical Service Name is the Critical Service Name

Repair Action

Check why Critical Service instances are restarting

Alert Rules

Alert Rule 1

This event will result in raising an alert for every **Unstable NF due to Frequent Restart** event generation based on realert configuration. The rising of this alert is for each unique value of **severity**, **criticalServiceName** and **networkFunctionName** fields with realert time counted separately.

Unstable NF due to Readiness Failure

Version

1.0

Description

Network Function is unstable due to Readiness Failure

Severity

- Critical

Auto Clearing

Yes

Clear Message

Critical Service:\$1 is stable.

Trigger Message

Due to Critical Service: **{Critical Service Name}** , NF: **{Network Function Name}** is unstable, timestamp: **{Event TimeStamp}**

where:

- Critical Service Name is the Critical Service Name
- Event TimeStamp is the Event Timestamp
- Network Function Name is the Network Function Name where:
- Critical Service Name is the Critical Service Name

Repair Action

Check why there are Readiness Probe failures for the Critical Service

Alert Rules

Alert Rule 1

This event will result in raising an alert for every **Unstable NF due to Readiness Failure** event generation based on realert configuration. The rising of this alert is for each unique value of **severity**, **criticalServiceName** and **networkFunctionName** fields with realert time counted separately.

Table of Contents

- [PFCP Association Status](#)
 - [PFCP Heart Beat Status](#)
-

PFCP Association Status

Version

1.0

Description

This event indicates that an association between two peers is going down or has gone down. When association is declared as down, no further sessions will be accepted towards that UPF, and the existing sessions on that association will be cleared. When the association is going down Warning log will be raised, when the association is declared as down, a Major event log will be raised

Severity

- Major
- Warning

Auto Clearing

Yes

Clear Message

PFCP Association is recovered between these peers, Local Node ID: \$1 | Remote Node ID: \$2 | Remote IP Address: \$3

Trigger Message

{Trigger} | PFCP Association is **{Association Status}** between these peers, Local Node ID: **{Local Node ID}** | Peer Node ID: **{Remote Node ID}** | Peer IP Address: **{Remote Node IP Address}**

where:

- Local Node ID is the PFCP Local Node ID of CPF/UPF
- Remote Node ID is the PFCP Peer Node ID of CPF/UPF
- Remote Node IP Address is the PFCP Peer Path IP Address of CPF/UPF
- Association Status is the Association Status, values are GOING_DOWN, DOWN
- Trigger is the Trigger reason for association status change where:
 - Local Node ID is the PFCP Local Node ID of CPF/UPF
 - Remote Node ID is the PFCP Peer Node ID of CPF/UPF
 - Remote Node IP Address is the PFCP Peer Path IP Address of CPF/UPF

Resource Xpath

Peer Identifier

The format of resource xpath is:

```
/pfcplocalpeer/[ {Local Node ID} ]/remotepeer/[ {Remote Node ID} ]
```

where:

- Local Node ID is the PFCP Local Node ID of CPF/UPF
- Remote Node ID is the PFCP Peer Node ID of CPF/UPF

Alert Rules

Alert Rule 1

This event will result in raising an alert for every **PFCP Association Status** event generation based on realert configuration.

Alert Rule 2

This event will result in raising an alert due to change with respect to **currentState** field based on realert configuration. The rising of this alert is for each unique value of **kubernetes.namespace_name** field with realert time counted separately.

Alert Rule 3

This event will result in raising an alert when **30** events are logged in **20** seconds time duration based on realert configuration.

Alert Rule 4

This event will result in raising an alert when **currentState** field are not set with any of the allowed values.

Allowed Values:

- **UP**

Alert Rule 5

This event will result in raising an alert when **currentState** and **severity** fields match with any of the disallowed values.

Disallowed Values:

- **DOWN**
- **GOINGDOWN**

PFCP Heart Beat Status

Version

1.0

Description

PFCP heart beat between peers has changed state. If the heartbeat is declared down for more than the configured pfc-heartbeat-down-threshold, the association will be restarted

Severity

- Major
- Warning

Auto Clearing

Yes

Clear Message

Heart Beat is recovered between these peers, Local Node ID: \$1 | Peer Node ID: \$2 | Peer IP Address: \$3

Trigger Message

Heart Beat is down/failed between these peers, Local Node ID: **{Local Node ID}** | Peer Node ID: **{Remote Node ID}** | Peer IP Address: **{Heart Beat IP Address}**

where:

- Local Node ID is the PFCP Entity Local Node ID of UPF/CPF
- Remote Node ID is the PFCP Peer Remote Node ID of UPF/CPF
- Heart Beat IP Address is the PFCP Path IP Address of UPF/CPF where:
- Local Node ID is the PFCP Entity Local Node ID of UPF/CPF
- Remote Node ID is the PFCP Peer Remote Node ID of UPF/CPF
- Heart Beat IP Address is the PFCP Path IP Address of UPF/CPF

Resource Xpath

Peer Identifier

The format of resource xpath is:

```
/pfcplocalpeer/[ {Local Node ID} ]/remoteppeer/[ {Remote Node ID} ]
```

where:

- Local Node ID is the PFCP Entity Local Node ID of UPF/CPF
- Remote Node ID is the PFCP Peer Remote Node ID of UPF/CPF

Alert Rules

Alert Rule 1

This event will result in raising an alert for every **PFCP Heart Beat Status** event generation based on realert configuration.

Alert Rule 2

This event will result in raising an alert due to change with respect to **currentState** field based on realert configuration. The rising of this alert is for each unique value of **kubernetes.namespace_name** field with realert time counted separately.

Alert Rule 3

This event will result in raising an alert when **30** events are logged in **20** seconds time duration based on realert configuration.

Alert Rule 4

This event will result in raising an alert when **currentState** field are not set with any of the allowed values.

Allowed Values:

- **UP**

Alert Rule 5

This event will result in raising an alert when **currentState** and **severity** fields match with any of the disallowed values.

Disallowed Values:

- **DOWN**
- **UNKNOWN**

Table of Contents

- [Kafka Broker Connection Status](#)
-

Kafka Broker Connection Status

Version

1.0

Description

This event is generated when the connection status towards Kafka broker changes. A critical event is raised when the Kafka broker connection goes down. The event is cleared when the connection is restored.

Severity

- Critical

Auto Clearing

Yes

Clear Message

Connection to Kafka broker \$1 is up

Trigger Message

Connection to Kafka broker **{Kafka Broker IP And Port}** is down

where:

- Kafka Broker IP And Port is the Configured Kafka broker endpoint in the format IP:port or FQDN:port. If port is not provided, a default port number 9092 is assumed. where:
- Kafka Broker IP And Port is the Configured Kafka broker endpoint in the format IP:port or FQDN:port. If port is not provided, a default port number 9092 is assumed.

Potential Impact

Applications will not be able to produce/consume the messages from Kafka Broker. This could have an impact based on the use cases like EDR generation, storing charging records, streaming packets etc.

Repair Action

Make sure the Kafka broker IP and port are configured correctly and the broker is reachable.

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- [Consul Read Status](#)
 - [Critical Service not Available](#)
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 - [LI PPE Destination Issue](#)
 - [NF Management Status](#)
 - [Unstable NF due to Frequent Restart](#)
 - [Unstable NF due to Readiness Failure](#)
-

Consul Read Status

Version

1.0

Description

Reading Previously Stored Status from Consul.

Severity

- Info

Auto Clearing

Yes

Clear Message

Consul Read Response:\$1 at:\$2 NF:\$4 is using the :\$3 InstanceID

Trigger Message

Consul Read Response: **{Consul Response}** at: **{Event TimeStamp}** NF: **{Network Function Name}** is using the : **{Info Message}** InstanceID

where:

- Consul Response is the Consul Read HTTP response
- Event TimeStamp is the Event Timestamp
- Info Message is the Info about the InstanceID (static/dynamic/existing)
- Network Function Name is the Network Function Name where:
- Consul Response is the Consul Read HTTP response
- Event TimeStamp is the Event Timestamp
- Info Message is the Info about the InstanceID (static/dynamic/existing)
- Network Function Name is the Network Function Name

Repair Action

Informative Event about the Consul Read Status, No Action Required

Alert Rules

Alert Rule 1

This event will result in raising an alert for every **Consul Read Status** event generation based on realert configuration. The rising of this alert is for each unique value of **severity**, **podName** and **networkFunctionName** fields with realert time counted separately.

Critical Service not Available

Version

1.0

Description

Critical Service is Unavailable

Severity

- Critical

Auto Clearing

Yes

Clear Message

Critical Service :\$1 has become Available (\$4) at :\$2 NF:\$3

Trigger Message

Critical Service : **{Critical Service Name}** is unavailable (**{Service Status}**) at : **{Event TimeStamp}** NF: **{Network Function Name}**

where:

- Critical Service Name is the Critical Service Name
- Event TimeStamp is the Event Timestamp
- Network Function Name is the Network Function Name
- Service Status is the Available/Required where:
 - Critical Service Name is the Critical Service Name
 - Event TimeStamp is the Event Timestamp
 - Network Function Name is the Network Function Name
 - Service Status is the Available/Required

Repair Action

Check if all the required critical services are available

Alert Rules

Alert Rule 1

This event will result in raising an alert for every **Critical Service not Available** event generation based on realert configuration. The rising of this alert is for each unique value of **severity**, **criticalServiceName** and **networkFunctionName** fields with realert time counted separately.

In Memory Buffer Full

Version

1.0

Description

This event indicates that in-memory buffer has become full(alarm set) or become half(alarm cleared) for a given buffer ID

Severity

- Critical

Auto Clearing

Yes

Clear Message

On Pod instance \$2 for a buffer with ID = \$1, in-memory buffer has become half, max buffer size in KB = \$3

Trigger Message

On Pod instance **{UniquePodId}** for a buffer with ID = **{BufferId}** , in-memory buffer has become full, max buffer size in KB = **{MaxBufferSize}**

where:

- BufferId is the Buffer Identifier for which the buffering is applicable
- UniquePodId is the Unique identifier for the pod that has generated this alert
- MaxBufferSize is the Maximum size of this buffer in KB where:
- BufferId is the Buffer Identifier for which the buffering is applicable
- UniquePodId is the Unique identifier for the pod that has generated this alert
- MaxBufferSize is the Maximum size of this buffer in KB

Potential Impact

Some PDUs being dropped from buffer with ID = \$1

Repair Action

Check if there is memory available on the pod and increase the size of the buffer

Alert Rules

Alert Rule 1

This event will result in raising an alert for every **In Memory Buffer Full** event generation based on realert configuration. The rising of this alert is for each unique value of **bufferid**, **uniquepodid** and **maxbuffersize** fields with realert time counted separately.

Kafka Broker Connection Status

Version

1.0

Description

This event is generated when the connection status towards Kafka broker changes. A critical event is raised when the Kafka broker connection goes down. The event is cleared when the connection is restored.

Severity

- Critical

Auto Clearing

Yes

Clear Message

Connection to Kafka broker \$1 is up

Trigger Message

Connection to Kafka broker **{Kafka Broker IP And Port}** is down

where:

- Kafka Broker IP And Port is the Configured Kafka broker endpoint in the format IP:port or FQDN:port. If port is not provided, a default port number 9092 is assumed. where:
- Kafka Broker IP And Port is the Configured Kafka broker endpoint in the format IP:port or FQDN:port. If port is not provided, a default port number 9092 is assumed.

Potential Impact

Applications will not be able to produce/consume the messages from Kafka Broker. This could have an impact based on the use cases like EDR generation, storing charging records, streaming packets etc.

Repair Action

Make sure the Kafka broker IP and port are configured correctly and the broker is reachable.

LI PPE Destination Issue

Version

1.0

Description

This event indicates that a destination issue has either occurred or cleared.

Severity

- Critical

Auto Clearing

Yes

Clear Message

On Pod instance \$3 for a DID \$1, Fault-Type \$2 cleared and Description: \$4.

Trigger Message

On Pod instance **{UniquePodId}** for a DID **{DID}** , Fault-Type **{FaultType}** occurred and Description: **{Description}** .

where:

- DID is the Destination Identifier in UUID format, configured either locally via cfgmgr pod or provisioned via LIPF
- FaultType is the Whether it is Warning, Terminating or Non-Terminating fault
- UniquePodId is the Unique identifier for the PPE pod that has generated this fault
- Description is the Human readable description of the fault, i.e. reason for this fault where:
- DID is the Destination Identifier in UUID format, configured either locally via cfgmgr pod or provisioned via LIPF
- FaultType is the Whether it is Warning, Terminating or Non-Terminating fault
- UniquePodId is the Unique identifier for the PPE pod that has generated this fault
- Description is the Human readable description of the fault, i.e. reason for this fault

Potential Impact

X2/X3 PDUs not being sent towards the MDF associated with the DID above.

Repair Action

Check if the MDF is up and running. If it is, check the connectivity from PPE to MDF. Also check if the destination port on MDF is open.

Alert Rules

Alert Rule 1

This event will result in raising an alert for every **LI PPE Destination Issue** event generation based on realert configuration. The rising of this alert is for each unique value of **did**, **faulttype**, **uniquepodid** and **description** fields with realert time counted separately.

NF Management Status

Version

1.0

Description

NF Status towards NRF

Severity

- Info

Auto Clearing

Yes

Clear Message

NF:\$4 has updated NRF with status:\$1 at:\$2 InstanceID:\$3

Trigger Message

NF: **{Network Function Name}** has updated NRF with status: **{NfMgmt Status}** at: **{Event TimeStamp}**
InstanceID: **{NF Instance ID}**

where:

- NfMgmt Status is the Nf Management Interface Status(Registered/DeRegistered/Maintenance)
- Event TimeStamp is the Event Timestamp
- NF Instance ID is the Network Function Instance ID
- Network Function Name is the Network Function Name where:
- NfMgmt Status is the Nf Management Interface Status(Registered/DeRegistered/Maintenance)
- Event TimeStamp is the Event Timestamp
- NF Instance ID is the Network Function Instance ID
- Network Function Name is the Network Function Name

Repair Action

Informative Event about the NF Status, No Action Required

Alert Rules

Alert Rule 1

This event will result in raising an alert for every **NF Management Status** event generation based on realert configuration. The rising of this alert is for each unique value of **networkFunctionName**, *** with realert time counted separately.

Unstable NF due to Frequent Restart

Version

1.0

Description

Network Function is unstable due to Frequent Restarts

Severity

- Critical

Auto Clearing

Yes

Clear Message

Critical Service:\$1 is stable.

Trigger Message

Due to Critical Service: **{Critical Service Name}** , NF: **{Network Function Name}** is unstable, timestamp: **{Event TimeStamp}**

where:

- Critical Service Name is the Critical Service Name
- Event TimeStamp is the Event Timestamp
- Network Function Name is the Network Function Name where:
- Critical Service Name is the Critical Service Name

Repair Action

Check why Critical Service instances are restarting

Alert Rules

Alert Rule 1

This event will result in raising an alert for every **Unstable NF due to Frequent Restart** event generation based on realert configuration. The rising of this alert is for each unique value of **severity**, **criticalServiceName** and **networkFunctionName** fields with realert time counted separately.

Unstable NF due to Readiness Failure

Version

1.0

Description

Network Function is unstable due to Readiness Failure

Severity

- Critical

Auto Clearing

Yes

Clear Message

Critical Service:\$1 is stable.

Trigger Message

Due to Critical Service: **{Critical Service Name}** , NF: **{Network Function Name}** is unstable, timestamp: **{Event TimeStamp}**

where:

- Critical Service Name is the Critical Service Name
- Event TimeStamp is the Event Timestamp
- Network Function Name is the Network Function Name where:
- Critical Service Name is the Critical Service Name

Repair Action

Check why there are Readiness Probe failures for the Critical Service

Alert Rules

Alert Rule 1

This event will result in raising an alert for every **Unstable NF due to Readiness Failure** event generation based on realert configuration. The rising of this alert is for each unique value of **severity**, **criticalServiceName** and **networkFunctionName** fields with realert time counted separately.

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- [Kafka Broker Connection Status](#)
 - [Tdef Proxy CPU Overload Major Alarm](#)
 - [Tdef Proxy CPU Overload Minor Alarm](#)
 - [Tdef Proxy CPU Overload Warning Alarm](#)
-

Kafka Broker Connection Status

Version

1.0

Description

This event is generated when the connection status towards Kafka broker changes. A critical event is raised when the Kafka broker connection goes down. The event is cleared when the connection is restored.

Severity

- Critical

Auto Clearing

Yes

Clear Message

Connection to Kafka broker \$1 is up

Trigger Message

Connection to Kafka broker **{Kafka Broker IP And Port}** is down

where:

- Kafka Broker IP And Port is the Configured Kafka broker endpoint in the format IP:port or FQDN:port. If port is not provided, a default port number 9092 is assumed. where:
- Kafka Broker IP And Port is the Configured Kafka broker endpoint in the format IP:port or FQDN:port. If port is not provided, a default port number 9092 is assumed.

Potential Impact

Applications will not be able to produce/consume the messages from Kafka Broker. This could have an impact based on the use cases like EDR generation, storing charging records, streaming packets etc.

Repair Action

Make sure the Kafka broker IP and port are configured correctly and the broker is reachable.

Tdef Proxy CPU Overload Major Alarm

Version

1.0

Description

Proxy CPU overload major alarm.

Severity

- Major

Auto Clearing

Yes

Clear Message

Proxy CPU usage level falls below the critical overloading threshold

Trigger Message

Proxy CPU usage level passes the critical overloading threshold.

Alert Rules

Alert Rule 1

This event will result in raising an alert for every **Tdef Proxy CPU Overload Major Alarm** event generation based on realert configuration. The rising of this alert is for each unique value of **podName** and **proxyId** fields with realert time counted separately.

Tdef Proxy CPU Overload Minor Alarm

Version

1.0

Description

Proxy CPU overload minor alarm.

Severity

- Minor

Auto Clearing

Yes

Clear Message

Proxy CPU usage level falls below the major overloading threshold

Trigger Message

Proxy CPU usage level passes the major overloading threshold.

Alert Rules

Alert Rule 1

This event will result in raising an alert for every **Tdef Proxy CPU Overload Minor Alarm** event generation based on realert configuration. The rising of this alert is for each unique value of **podName** and **proxyId** fields with realert time counted separately.

Tdef Proxy CPU Overload Warning Alarm

Version

1.0

Description

Proxy CPU overload warning alarm.

Severity

- Warning

Auto Clearing

Yes

Clear Message

Proxy CPU usage level falls below the minor overloading threshold

Trigger Message

Proxy CPU usage level passes the minor overloading threshold.

Alert Rules

Alert Rule 1

This event will result in raising an alert for every **Tdef Proxy CPU Overload Warning Alarm** event generation based on realert configuration. The rising of this alert is for each unique value of **podName** and **proxyId** fields with realert time counted separately.

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- [Redis Failover Election Won Event](#)
 - [Redis Liveness Failure Event](#)
 - [Redis Readiness Failure Event](#)
-

Redis Failover Election Won Event

Version

1.0

Description

This event indicates the elections happening in replica node after primary becomes unavailable for a configurable period of `clusterNodeTimeout`, which is set to 15ms

Severity

- Major

Auto Clearing

No

Root Cause

Reasons could be that a primary becomes unavailable due to network issue or a pod restart due to some failure in that failure domain.

Trigger Message

Failover election won: I'm the new master

Potential Impact

Primary node gets changed and role gets shifted to one of its replica

Repair Action

Verify the data and slots are in perfect sync with new primary or not

Alert Rules

Alert Rule 1

This event will result in raising an alert for every **Redis Failover Election Won Event** event generation based on realert configuration. The rising of this alert is for each unique value of **kubernetes.pod_name**, *** with realert time counted separately.

Redis Liveness Failure Event

Version

1.0

Description

Redis node is not responsive to liveness probe and might not be able to serve the requests

Severity

- Critical

Auto Clearing

No

Root Cause

This can happen when for any reason a redis-server process is not running or not able to serve requests. This alert can be seen when cluster is balancing topology or when a pod is being restarted.

Trigger Message

liveness check failed, err:Liveness failed, redis node unresponsive

Potential Impact

Redis server may be unstable and unable to serve requests if node keeps bein unresponsive.

Repair Action

Check the resource usage for redis pods and restart the pod if needed

Alert Rules

Alert Rule 1

This event will result in raising an alert for every **Redis Liveness Failure Event** event generation based on realert configuration. The rising of this alert is for each unique value of **kubernetes.pod_name**, *** with realert time counted separately.

Redis Readiness Failure Event

Version

1.0

Description

Redis hash slots could not be assigned due to some redis cluster issue/ this pod was unable to join the redis cluster bus

Severity

- Critical

Auto Clearing

No

Root Cause

Reason could be hard reboot of worker nodes resulting in readiness probe failure

Trigger Message

readiness check failed, err:Readiness failed, cluster slots response empty

Potential Impact

Redis server may be unstable if lot of redis nodes lose slots.

Repair Action

Reinstall redis cluster manually by uninstalling chart /deleting pods

Alert Rules

Alert Rule 1

This event will result in raising an alert for every **Redis Readiness Failure Event** event generation based on realert configuration. The rising of this alert is for each unique value of **kubernetes.pod_name**, *** with realert time counted separately.

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- [REDIS Nodes Unavailable Event](#)
-

REDIS Nodes Unavailable Event

Version

1.0

Description

Intended number of Redis pods are not active/up

Severity

- Major

Auto Clearing

No

Root Cause

Either Redis operator is unable to reconcile the cluster due to various reasons or pods are stuck in Terminating/Not Ready state etc

Trigger Message

not all redis nodes are running, numberOfPods: **{Number Of Pods}** , numberOfRedisNodesRunning: **{Number Of Redis Nodes Running}**

where:

- Number Of Pods is the Total number of pods required in Redis cluster.
- Number Of Redis Nodes Running is the Total number of pods running in a Redis cluster.

Potential Impact

Eventually Redis server may become unavailable

Repair Action

Check if Redis operator is able to reconcile the Redis cluster and Redis cluster should recover automatically. If not, manually re-install cluster again

Alert Rules

Alert Rule 1

This event will result in raising an alert for every **REDIS Nodes Unavailable Event** event generation based on realert configuration.

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- [Redis Geo Redundancy Connection Status Event](#)
-

Redis Geo Redundancy Connection Status Event

Version

1.0

Description

This event is generated when the connection status towards Remote Site goes down. A critical event is raised when the remote site connection goes down. The event is cleared when the connection is restored.

Severity

- Critical

Auto Clearing

Yes

Clear Message

Connection to Remote Redis Server \$2:\$1 is up

Trigger Message

Connection to Remote Redis Server **{Remote Node IP Address}** : **{Remote Site Port}** is down

where:

- Remote Site Port is the Configured Remote Site port
- Remote Node IP Address is the Configured Remote Site endpoint where:
- Remote Site Port is the Configured Remote Site port
- Remote Node IP Address is the Configured Remote Site endpoint

Potential Impact

Replication to remote site is broken as connection status is down which leads to disruption in the data replication.

Repair Action

Make sure the Remote Site IP and port are configured correctly and the site is reachable.

Alert Rules

Alert Rule 1

This event will result in raising an alert due to change with respect to **currentState** field based on realert configuration. The rising of this alert is for each unique value of **kubernetes.namespace_name** field with realert time counted separately.

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- [Consul Read Status](#)
- [Critical Service not Available](#)
- [License expired](#)
- [License limit reached](#)
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- [NF Management Status](#)
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- [SMF Critical Service Status Change Event](#)
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- [SMF PFCP Association Status Change Event](#)

- [Unstable NF due to Frequent Restart](#)
- [Unstable NF due to Readiness Failure](#)

Consul Read Status

Version

1.0

Description

Reading Previously Stored Status from Consul.

Severity

- Info

Auto Clearing

Yes

Clear Message

Consul Read Response:\$1 at:\$2 NF:\$4 is using the :\$3 InstanceID

Trigger Message

Consul Read Response: **{Consul Response}** at: **{Event TimeStamp}** NF: **{Network Function Name}** is using the : **{Info Message}** InstanceID

where:

- Consul Response is the Consul Read HTTP response
- Event TimeStamp is the Event Timestamp
- Info Message is the Info about the InstanceID (static/dynamic/existing)
- Network Function Name is the Network Function Name where:
- Consul Response is the Consul Read HTTP response
- Event TimeStamp is the Event Timestamp
- Info Message is the Info about the InstanceID (static/dynamic/existing)
- Network Function Name is the Network Function Name

Repair Action

Informative Event about the Consul Read Status, No Action Required

Alert Rules

Alert Rule 1

This event will result in raising an alert for every **Consul Read Status** event generation based on realert configuration. The rising of this alert is for each unique value of **severity**, **podName** and **networkFunctionName** fields with realert time counted separately.

Critical Service not Available

Version

1.0

Description

Critical Service is Unavailable

Severity

- Critical

Auto Clearing

Yes

Clear Message

Critical Service :\$1 has become Available (\$4) at :\$2 NF:\$3

Trigger Message

Critical Service : **{Critical Service Name}** is unavailable (**{Service Status}**) at : **{Event TimeStamp}** NF: **{Network Function Name}**

where:

- Critical Service Name is the Critical Service Name
- Event TimeStamp is the Event Timestamp
- Network Function Name is the Network Function Name
- Service Status is the Available/Required where:
- Critical Service Name is the Critical Service Name
- Event TimeStamp is the Event Timestamp
- Network Function Name is the Network Function Name
- Service Status is the Available/Required

Repair Action

Check if all the required critical services are available

Alert Rules

Alert Rule 1

This event will result in raising an alert for every **Critical Service not Available** event generation based on realert configuration. The rising of this alert is for each unique value of **severity**, **criticalServiceName** and **networkFunctionName** fields with realert time counted separately.

License expired

Version

1.0

Description

License expired.

Severity

- Info
- Minor
- Major
- Critical

Auto Clearing

Yes

Clear Message

License :\$1 expiration date :\$2

Trigger Message

License : **{License name version}** expiration date : **{License expiration date}**

where:

- License name version is the License name and version.
- License expiration date is the License expiration date. where:
- License name version is the License name and version.
- License expiration date is the License expiration date.

Repair Action

Check license expiration date.

License limit reached

Version

1.0

Description

License limit reachead.

Severity

- Info
- Minor
- Major
- Critical

Auto Clearing

Yes

Clear Message

License :\$1 limit cleared : number of SAUs \$4, entitlements used \$3, entitlements requested \$2

Trigger Message

License : **{License name version}** limit reached : number of SAUs **{Number of SAUs}** , entitlements used **{Number of entitlements requested}** , entitlements requested **{Number of entitlements used}**

where:

- License name version is the License name and version.
- Number of entitlements used is the Number of entitlements used.
- Number of entitlements requested is the Number of entitlements requested.
- Number of SAUs is the Number of SAUs. where:
- License name version is the License name and version.
- Number of entitlements used is the Number of entitlements used.
- Number of entitlements requested is the Number of entitlements requested.
- Number of SAUs is the Number of SAUs.

Repair Action

Check license soft limit.

License required

Version

1.0

Description

License is required.

Severity

- Critical
- Info

Auto Clearing

Yes

Clear Message

License :\$1 is obtained.

Trigger Message

License : **{License name version}** is required

where:

- License name version is the License name and version. where:
- License name version is the License name and version.

License server is not reachable

Version

1.0

Description

License server is not reachables.

Severity

- Critical
- Info

Auto Clearing

Yes

Clear Message

Connected to license server :\$1

Trigger Message

Failed to connect to license server : **{License server}**

where:

- License server is the License server name or ip address. where:
- License server is the License server name or ip address.

Potential Impact

Licenses cannot be obtained from the server.

Repair Action

Check license server connection.

NF Management Status

Version

1.0

Description

NF Status towards NRF

Severity

- Info

Auto Clearing

Yes

Clear Message

NF:\$4 has updated NRF with status:\$1 at:\$2 InstanceID:\$3

Trigger Message

NF: **{Network Function Name}** has updated NRF with status: **{NfMgmt Status}** at: **{Event TimeStamp}**
InstanceID: **{NF Instance ID}**

where:

- NfMgmt Status is the Nf Management Interface Status(Registered/DeRegistered/Maintenance)
- Event TimeStamp is the Event Timestamp
- NF Instance ID is the Network Function Instance ID
- Network Function Name is the Network Function Name where:
- NfMgmt Status is the Nf Management Interface Status(Registered/DeRegistered/Maintenance)
- Event TimeStamp is the Event Timestamp
- NF Instance ID is the Network Function Instance ID

- Network Function Name is the Network Function Name

Repair Action

Informative Event about the NF Status, No Action Required

Alert Rules

Alert Rule 1

This event will result in raising an alert for every **NF Management Status** event generation based on realert configuration. The rising of this alert is for each unique value of **networkFunctionName**, *** with realert time counted separately.

NF at Critical Load Level

Version

1.0

Description

Network Function is now Critically loaded, new sessions will get affected

Severity

- Critical

Auto Clearing

Yes

Clear Message

NF:\$1 no longer at Critical Load Levels.

Trigger Message

NF: **{Network Function Name}** is now running with Critical Load

where:

- Network Function Name is the Network Function Name where:
- Network Function Name is the Network Function Name

Repair Action

Check why NF is in Load, show smfmonitor nf-load command gives detailed info

Alert Rules

Alert Rule 1

This event will result in raising an alert for every **NF at Critical Load Level** event generation based on realert configuration. The rising of this alert is for each unique value of **severity** and **networkFunctionName** fields with realert time counted separately.

NF at Major Load Level

Version

1.0

Description

Network Function is now Majorly loaded, new sessions will start to get affected

Severity

- Major

Auto Clearing

Yes

Clear Message

NF:\$1 no longer Majorly Loaded.

Trigger Message

NF: **{Network Function Name}** is now running with Major Load

where:

- Network Function Name is the Network Function Name where:
- Network Function Name is the Network Function Name

Repair Action

Check why NF is in Load, show smfmonitor nf-load command gives detailed info

Alert Rules

Alert Rule 1

This event will result in raising an alert for every **NF at Major Load Level** event generation based on realert configuration. The rising of this alert is for each unique value of **severity** and **networkFunctionName** fields with realert time counted separately.

NF at Minor Load Level

Version

1.0

Description

Network Function is now Minorly loaded, new sessions may start to get affected

Severity

- Minor

Auto Clearing

Yes

Clear Message

NF:\$1 no longer Loaded.

Trigger Message

NF: **{Network Function Name}** is now running with Minor Load

where:

- Network Function Name is the Network Function Name where:
- Network Function Name is the Network Function Name

Repair Action

Check why NF is in Load, show smfmonitor nf-load command gives detailed info

Alert Rules

Alert Rule 1

This event will result in raising an alert for every **NF at Minor Load Level** event generation based on realert configuration. The rising of this alert is for each unique value of **severity** and **networkFunctionName** fields with realert time counted separately.

SMF Critical Service Status Change Event

Version

1.0

Description

This event indicates that a particular critical service, required for the SMF instance for its functionality, has become available or unavailable. When the critical service goes down, a warning event log is raised and when the critical service comes up, an info log event log will be raised.

Severity

- Info
- Warning

Auto Clearing

No

Trigger Message

Status of a critical service has changed. Service Name: **{Critical Service Name}** , Service Type: **{Critical Service Type}** , Service Criticality: **{Critical Service Criticality Type}** , Service Instances: **{Critical Service Instances Count}** , Service Status: **{Critical Service Status}**

where:

- Critical Service Name is the Name of the critical service for which the status has changed
- Critical Service Type is the Type of the critical service namely Application, Common and Paas
- Critical Service Criticality Type is the Criticality type of the critical service namely Ready and Continuity
- Critical Service Instances Count is the Number of instances of a critical service which determine its availability/unavailability to the SMF instance
- Critical Service Status is the Status of a critical service namely up and down

Potential Impact

Change in status up/down might have direct impact on the SMF instance registration/deregistration respectively.

Repair Action

Check if all the required critical services are available for the SMF instance.

SMF Invalid DNN Configured Event

Version

1.0

Description

This event indicates that a particular configured DNN is invalid and does not matches any of the PLMN configured.

Severity

- Warning

Auto Clearing

No

Trigger Message

Invalid DNN [**{DNN Name}**] configured, does not matches any of PLMN configured in SMF service construct

where:

- DNN Name is the Name of the invalid DNN configured

Potential Impact

This invalid DNN will not be included in the NF profile for registration with NRF.

Repair Action

Check and correc the Invalid DNN configured.

Alert Rules

Alert Rule 1

This event will result in raising an alert for every **SMF Invalid DNN Configured Event** event generation based on realert configuration. The rising of this alert is for each unique value of **severity** and **dnnName** fields with realert time counted separately.

SMF NFManagement Service Deregistration Event

Version

1.0

Description

This event indicates that the given SMF instance is deregistered with NRF due to either one of the critical services [Application/Common/Paas] or application configuration becomes unavailable due to which the SMF instance will not be able to provide its services.

Once the critical services or application configuration is restored resulting in the SMF instance registration with NRF, the event will be cleared.

Severity

- Critical

Auto Clearing

Yes

Clear Message

The SMF instance is in registered state with NRF now

Trigger Message

The SMF instance is deregistered with NRF. SMF Instance ID: ***{NF Instance ID}***

where:

- NF Instance ID is the UUID of the SMF Instance

Potential Impact

SMF instance will not be able to provide its services due to which there may be temporary service discontinuity

Repair Action

Check if all the required critical services are available for the SMF instance along with its configuration

SMF PFCP Association Status Change Event

Version

1.0

Description

This event indicates that all the PFCP Associations went down, impacting the service. It needs at least 1 active association for the SMF functionality.

Severity

- Critical

Auto Clearing

Yes

Clear Message

PFCP Associations has come up

Trigger Message

All of the PFCP Associations have gone Down

Potential Impact

Change in status up/down might have direct impact on the SMF instance registration status

Repair Action

Check the PFCP Association status'

Unstable NF due to Frequent Restart

Version

1.0

Description

Network Function is unstable due to Frequent Restarts

Severity

- Critical

Auto Clearing

Yes

Clear Message

Critical Service:\$1 is stable.

Trigger Message

Due to Critical Service: **{Critical Service Name}** , NF: **{Network Function Name}** is unstable, timestamp: **{Event TimeStamp}**

where:

- Critical Service Name is the Critical Service Name
- Event TimeStamp is the Event Timestamp
- Network Function Name is the Network Function Name where:
- Critical Service Name is the Critical Service Name

Repair Action

Check why Critical Service instances are restarting

Alert Rules

Alert Rule 1

This event will result in raising an alert for every **Unstable NF due to Frequent Restart** event generation based on realert configuration. The rising of this alert is for each unique value of **severity**, **criticalServiceName** and **networkFunctionName** fields with realert time counted separately.

Unstable NF due to Readiness Failure

Version

1.0

Description

Network Function is unstable due to Readiness Failure

Severity

- Critical

Auto Clearing

Yes

Clear Message

Critical Service:\$1 is stable.

Trigger Message

Due to Critical Service: **{Critical Service Name}** , NF: **{Network Function Name}** is unstable, timestamp: **{Event TimeStamp}**

where:

- Critical Service Name is the Critical Service Name
- Event TimeStamp is the Event Timestamp
- Network Function Name is the Network Function Name where:
- Critical Service Name is the Critical Service Name

Repair Action

Check why there are Readiness Probe failures for the Critical Service

Alert Rules

Alert Rule 1

This event will result in raising an alert for every **Unstable NF due to Readiness Failure** event generation based on realert configuration. The rising of this alert is for each unique value of **severity**, **criticalServiceName** and **networkFunctionName** fields with realert time counted separately.

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Consul Read Status

Version

1.0

Description

Reading Previously Stored Status from Consul.

Severity

- Info

Auto Clearing

Yes

Clear Message

Consul Read Response:\$1 at:\$2 NF:\$4 is using the :\$3 InstanceID

Trigger Message

Consul Read Response: **{Consul Response}** at: **{Event TimeStamp}** NF: **{Network Function Name}** is using the : **{Info Message}** InstanceID

where:

- Consul Response is the Consul Read HTTP response
- Event TimeStamp is the Event Timestamp
- Info Message is the Info about the InstanceID (static/dynamic/existing)
- Network Function Name is the Network Function Name where:
- Consul Response is the Consul Read HTTP response
- Event TimeStamp is the Event Timestamp
- Info Message is the Info about the InstanceID (static/dynamic/existing)
- Network Function Name is the Network Function Name

Repair Action

Informative Event about the Consul Read Status, No Action Required

Alert Rules

Alert Rule 1

This event will result in raising an alert for every **Consul Read Status** event generation based on realert configuration. The rising of this alert is for each unique value of **severity**, **podName** and **networkFunctionName** fields with realert time counted separately.

Critical Service not Available

Version

1.0

Description

Critical Service is Unavailable

Severity

- Critical

Auto Clearing

Yes

Clear Message

Critical Service :\$1 has become Available (\$4) at :\$2 NF:\$3

Trigger Message

Critical Service : **{Critical Service Name}** is unavailable (**{Service Status}**) at : **{Event TimeStamp}** NF: **{Network Function Name}**

where:

- Critical Service Name is the Critical Service Name
- Event TimeStamp is the Event Timestamp
- Network Function Name is the Network Function Name
- Service Status is the Available/Required where:
 - Critical Service Name is the Critical Service Name
 - Event TimeStamp is the Event Timestamp
 - Network Function Name is the Network Function Name
 - Service Status is the Available/Required

Repair Action

Check if all the required critical services are available

Alert Rules

Alert Rule 1

This event will result in raising an alert for every **Critical Service not Available** event generation based on realert configuration. The rising of this alert is for each unique value of **severity**, **criticalServiceName** and **networkFunctionName** fields with realert time counted separately.

In Memory Buffer Full

Version

1.0

Description

This event indicates that in-memory buffer has become full(alarm set) or become half(alarm cleared) for a given buffer ID

Severity

- Critical

Auto Clearing

Yes

Clear Message

On Pod instance \$2 for a buffer with ID = \$1, in-memory buffer has become half, max buffer size in KB = \$3

Trigger Message

On Pod instance **{UniquePodId}** for a buffer with ID = **{BufferId}** , in-memory buffer has become full, max buffer size in KB = **{MaxBufferSize}**

where:

- BufferId is the Buffer Identifier for which the buffering is applicable
- UniquePodId is the Unique identifier for the pod that has generated this alert
- MaxBufferSize is the Maximum size of this buffer in KB where:
- BufferId is the Buffer Identifier for which the buffering is applicable
- UniquePodId is the Unique identifier for the pod that has generated this alert
- MaxBufferSize is the Maximum size of this buffer in KB

Potential Impact

Some PDUs being dropped from buffer with ID = \$1

Repair Action

Check if there is memory available on the pod and increase the size of the buffer

Alert Rules

Alert Rule 1

This event will result in raising an alert for every **In Memory Buffer Full** event generation based on realert configuration. The rising of this alert is for each unique value of **bufferid**, **uniquepodid** and **maxbuffersize** fields with realert time counted separately.

Kafka Broker Connection Status

Version

1.0

Description

This event is generated when the connection status towards Kafka broker changes. A critical event is raised when the Kafka broker connection goes down. The event is cleared when the connection is restored.

Severity

- Critical

Auto Clearing

Yes

Clear Message

Connection to Kafka broker \$1 is up

Trigger Message

Connection to Kafka broker **{Kafka Broker IP And Port}** is down

where:

- Kafka Broker IP And Port is the Configured Kafka broker endpoint in the format IP:port or FQDN:port. If port is not provided, a default port number 9092 is assumed. where:
- Kafka Broker IP And Port is the Configured Kafka broker endpoint in the format IP:port or FQDN:port. If port is not provided, a default port number 9092 is assumed.

Potential Impact

Applications will not be able to produce/consume the messages from Kafka Broker. This could have an impact based on the use cases like EDR generation, storing charging records, streaming packets etc.

Repair Action

Make sure the Kafka broker IP and port are configured correctly and the broker is reachable.

Missing Network Instance Profile Analyzer Instance Config

Version

1.0

Description

This event indicates that Network Instance Profile Subscriber Analyzer Instance value not configured in SMF. PDU Session establishment will fail without network instance profile.

Severity

- Warning

Auto Clearing

No

Trigger Message

Missing Network Instance Profile Analyzer Instance Config

Potential Impact

PDU Session establishment will fail

Repair Action

Configure network instance profile in SMF config manager

Alert Rules

Alert Rule 1

This event will result in raising an alert for every **Missing Network Instance Profile Analyzer Instance Config** event generation based on realert configuration.

Missing Or Invalid Network Instance Profile

Version

1.0

Description

This event indicates that Network Instance Profile not configured in SMF. Or N3/N6/N9 network instance values not configured in network instance profile. PDU Session establishment will fail without network instance profile.

Severity

- Warning

Auto Clearing

No

Trigger Message

Missing Network Instance Profile Or N3/N6/N9 network instance values not configured in profile:
{nwinstanceprofilename}

where:

- nwinstanceprofilename is the Network Instance Profile Name

Potential Impact

PDU Session establishment will fail

Repair Action

Configure network instance profile with valid N3/N6/N9 network instance values in SMF config manager

Alert Rules

Alert Rule 1

This event will result in raising an alert for every **Missing Or Invalid Network Instance Profile** event generation based on realert configuration.

Missing SSC Mode In UDM Subscriber Info

Version

1.0

Description

This event indicates that UDM has not provided SSC Mode in default SSC mode or allowed SSC Mode attribute.

SMF shall use SSC mode configured in SMF Subscriber Management Profile.

Severity

- Info

Auto Clearing

No

Trigger Message

Missing SSC Mode Information from UDM for SUPI: **{supi}**

where:

- supi is the UE SUPI Information

Potential Impact

Not using UDM subscribed SSC Mode for PDU session

Repair Action

Configure SSC Mode in UDM and UDM shall send SSC Mode to SMF while quering subscriber information

NF Management Status

Version

1.0

Description

NF Status towards NRF

Severity

- Info

Auto Clearing

Yes

Clear Message

NF:\$4 has updated NRF with status:\$1 at:\$2 InstanceID:\$3

Trigger Message

NF: **{Network Function Name}** has updated NRF with status: **{NfMgmt Status}** at: **{Event TimeStamp}**
InstanceID: **{NF Instance ID}**

where:

- NfMgmt Status is the Nf Management Interface Status(Registered/DeRegistered/Maintenance)
- Event TimeStamp is the Event Timestamp
- NF Instance ID is the Network Function Instance ID
- Network Function Name is the Network Function Name where:
- NfMgmt Status is the Nf Management Interface Status(Registered/DeRegistered/Maintenance)
- Event TimeStamp is the Event Timestamp
- NF Instance ID is the Network Function Instance ID
- Network Function Name is the Network Function Name

Repair Action

Informative Event about the NF Status, No Action Required

Alert Rules

Alert Rule 1

This event will result in raising an alert for every **NF Management Status** event generation based on realert configuration. The rising of this alert is for each unique value of **networkFunctionName**, *** with realert time counted separately.

Online Charging Attributes Not Present In Charging Profile

Version

1.0

Description

This event indicates that Online charging attributes are not present in charging profile.

SMF could not allocate default quota for failed charging session and terminating the pdu session

Severity

- Info

Auto Clearing

No

Trigger Message

Online charging attribute not present in charging profile **{charging_profile_name}** . Terminating the session for supi: **{supi}** , dnn: **{dnn}** , snssai: **{snssai}**

where:

- charging_profile_name is the Charging profile associated with the pdusession
- supi is the UE SUPI Information
- dnn is the DNN of pdu session
- snssai is the SNSSAI of pdu session

Potential Impact

PDU Session establishment/update will fail

Repair Action

Configure online charging attributes under charging profile

Session Level Triggers Not Present In Charging Profile

Version

1.0

Description

This event indicates that, if chf session fails and smf is configured to continue the session, it need atleast session level trigger to continue the session.

Since session level triggers are not present terminating the session

Severity

- Info

Auto Clearing

No

Trigger Message

Session level triggers not present in charging profile **{charging_profile_name}** . Could not continue the failed charging session and terminating the pdu session for supi: **{supi}** , dnn: **{dnn}** , snssai: **{snssai}**

where:

- charging_profile_name is the Charging profile associated with the pdusession
- supi is the UE SUPI Information
- dnn is the DNN of pdu session
- snssai is the SNSSAI of pdu session

Potential Impact

PDU Session establishment/update will fail

Repair Action

Configure session level trigger under charging profile

Unstable NF due to Frequent Restart

Version

1.0

Description

Network Function is unstable due to Frequent Restarts

Severity

- Critical

Auto Clearing

Yes

Clear Message

Critical Service:\$1 is stable.

Trigger Message

Due to Critical Service: **{Critical Service Name}** , NF: **{Network Function Name}** is unstable, timestamp: **{Event TimeStamp}**

where:

- Critical Service Name is the Critical Service Name
- Event TimeStamp is the Event Timestamp
- Network Function Name is the Network Function Name where:
- Critical Service Name is the Critical Service Name

Repair Action

Check why Critical Service instances are restarting

Alert Rules

Alert Rule 1

This event will result in raising an alert for every **Unstable NF due to Frequent Restart** event generation based on realert configuration. The rising of this alert is for each unique value of **severity**, **criticalServiceName** and **networkFunctionName** fields with realert time counted separately.

Unstable NF due to Readiness Failure

Version

1.0

Description

Network Function is unstable due to Readiness Failure

Severity

- Critical

Auto Clearing

Yes

Clear Message

Critical Service:\$1 is stable.

Trigger Message

Due to Critical Service: **{Critical Service Name}** , NF: **{Network Function Name}** is unstable, timestamp: **{Event TimeStamp}**

where:

- Critical Service Name is the Critical Service Name
- Event TimeStamp is the Event Timestamp
- Network Function Name is the Network Function Name where:
- Critical Service Name is the Critical Service Name

Repair Action

Check why there are Readiness Probe failures for the Critical Service

Alert Rules

Alert Rule 1

This event will result in raising an alert for every **Unstable NF due to Readiness Failure** event generation based on realert configuration. The rising of this alert is for each unique value of **severity**, **criticalServiceName** and **networkFunctionName** fields with realert time counted separately.

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UDSF Association Status

Version

1.0

Description

A database connection is not establishing.

Severity

- Warning

Auto Clearing

Yes

Clear Message

The database is operating as expected on behalf of federation namespace \$1

Trigger Message

A error has been detected in the database on behalf of federation namespace ***{Federation Namespace}*** : ***{DbError}***

where:

- Federation Namespace is the Name of the federation namespace that detected the failure event
- DbError is the DbError as reported by the database where:
- Federation Namespace is the Name of the federation namespace that detected the failure event

Potential Impact

The database may have stopped and may be unable to provide service.

Repair Action

Diagnose the database error given in the information. It may be necessary to redeploy the database to recover.

Alert Rules

Alert Rule 1

This event will result in raising an alert due to change with respect to ***severity*** field based on realert configuration. The rising of this alert is for each unique value of ***federationNamespace*** field with realert time counted separately.

UDSF Auditing Failed

Version

1.0

Description

Unable to fetch the required keys from database for auditing.

Severity

- Critical

Auto Clearing

Yes

Clear Message

The database is operating as expected on behalf of federation namespace \$1

Trigger Message

A error has been detected in the database on behalf of federation namespace ***{Federation Namespace}*** : ***{DbError}***

where:

- Federation Namespace is the Name of the federation namespace that detected the failure event
- DbError is the DbError as reported by the database where:
- Federation Namespace is the Name of the federation namespace that detected the failure event

Potential Impact

The database may have stopped and may be unable to provide service.

Repair Action

Diagnose the database error given in the information. It may be necessary to redeploy the database to recover.

Alert Rules

Alert Rule 1

This event will result in raising an alert due to change with respect to ***severity*** field based on realert configuration. The rising of this alert is for each unique value of ***federationNamespace*** field with realert time counted separately.

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Consul Read Status

Version

1.0

Description

Reading Previously Stored Status from Consul.

Severity

- Info

Auto Clearing

Yes

Clear Message

Consul Read Response:\$1 at:\$2 NF:\$4 is using the :\$3 InstanceID

Trigger Message

Consul Read Response: **{Consul Response}** at: **{Event TimeStamp}** NF: **{Network Function Name}** is using the : **{Info Message}** InstanceID

where:

- Consul Response is the Consul Read HTTP response
- Event TimeStamp is the Event Timestamp
- Info Message is the Info about the InstanceID (static/dynamic/existing)
- Network Function Name is the Network Function Name where:
- Consul Response is the Consul Read HTTP response
- Event TimeStamp is the Event Timestamp
- Info Message is the Info about the InstanceID (static/dynamic/existing)
- Network Function Name is the Network Function Name

Repair Action

Informative Event about the Consul Read Status, No Action Required

Alert Rules

Alert Rule 1

This event will result in raising an alert for every **Consul Read Status** event generation based on realert configuration. The rising of this alert is for each unique value of **severity**, **podName** and **networkFunctionName** fields with realert time counted separately.

Critical Service not Available

Version

1.0

Description

Critical Service is Unavailable

Severity

- Critical

Auto Clearing

Yes

Clear Message

Critical Service :\$1 has become Available (\$4) at :\$2 NF:\$3

Trigger Message

Critical Service : **{Critical Service Name}** is unavailable (**{Service Status}**) at : **{Event TimeStamp}** NF: **{Network Function Name}**

where:

- Critical Service Name is the Critical Service Name
- Event TimeStamp is the Event Timestamp
- Network Function Name is the Network Function Name
- Service Status is the Available/Required where:
- Critical Service Name is the Critical Service Name
- Event TimeStamp is the Event Timestamp
- Network Function Name is the Network Function Name
- Service Status is the Available/Required

Repair Action

Check if all the required critical services are available

Alert Rules

Alert Rule 1

This event will result in raising an alert for every **Critical Service not Available** event generation based on realert configuration. The rising of this alert is for each unique value of **severity**, **criticalServiceName** and **networkFunctionName** fields with realert time counted separately.

License expired

Version

1.0

Description

License expired.

Severity

- Info
- Minor
- Major
- Critical

Auto Clearing

Yes

Clear Message

License :\$1 expiration date :\$2

Trigger Message

License : **{License name version}** expiration date : **{License expiration date}**

where:

- License name version is the License name and version.
- License expiration date is the License expiration date. where:
- License name version is the License name and version.
- License expiration date is the License expiration date.

Repair Action

Check license expiration date.

License limit reached

Version

1.0

Description

License limit reachhead.

Severity

- Info

- Minor
- Major
- Critical

Auto Clearing

Yes

Clear Message

License :\$1 limit cleared : number of SAUs \$4, entitlements used \$3, entitlements requested \$2

Trigger Message

License : **{License name version}** limit reached : number of SAUs **{Number of SAUs}** , entitlements used **{Number of entitlements requested}** , entitlements requested **{Number of entitlements used}**

where:

- License name version is the License name and version.
- Number of entitlements used is the Number of entitlements used.
- Number of entitlements requested is the Number of entitlements requested.
- Number of SAUs is the Number of SAUs. where:
- License name version is the License name and version.
- Number of entitlements used is the Number of entitlements used.
- Number of entitlements requested is the Number of entitlements requested.
- Number of SAUs is the Number of SAUs.

Repair Action

Check license soft limit.

License required

Version

1.0

Description

License is required.

Severity

- Critical
- Info

Auto Clearing

Yes

Clear Message

License :\$1 is obtained.

Trigger Message

License : **{License name version}** is required

where:

- License name version is the License name and version. where:
- License name version is the License name and version.

License server is not reachable

Version

1.0

Description

License server is not reachables.

Severity

- Critical
- Info

Auto Clearing

Yes

Clear Message

Connected to license server :\$1

Trigger Message

Failed to connect to license server : **{License server}**

where:

- License server is the License server name or ip address. where:
- License server is the License server name or ip address.

Potential Impact

Licenses cannot be obtained from the server.

Repair Action

Check license server connection.

NF Management Status

Version

1.0

Description

NF Status towards NRF

Severity

- Info

Auto Clearing

Yes

Clear Message

NF:\$4 has updated NRF with status:\$1 at:\$2 InstanceID:\$3

Trigger Message

NF: **{Network Function Name}** has updated NRF with status: **{NfMgmt Status}** at: **{Event TimeStamp}**
InstanceID: **{NF Instance ID}**

where:

- NfMgmt Status is the Nf Management Interface Status(Registered/DeRegistered/Maintenance)
- Event TimeStamp is the Event Timestamp
- NF Instance ID is the Network Function Instance ID
- Network Function Name is the Network Function Name where:
- NfMgmt Status is the Nf Management Interface Status(Registered/DeRegistered/Maintenance)
- Event TimeStamp is the Event Timestamp
- NF Instance ID is the Network Function Instance ID
- Network Function Name is the Network Function Name

Repair Action

Informative Event about the NF Status, No Action Required

Alert Rules

Alert Rule 1

This event will result in raising an alert for every **NF Management Status** event generation based on realert configuration. The rising of this alert is for each unique value of **networkFunctionName**, *** with realert time counted separately.

UPF Critical Service Status Change Event

Version

1.0

Description

This event indicates that a particular critical service, required for the UPF instance for its functionality, has become available or unavailable. When the critical service goes down, a warning event log is raised and when the critical service comes up, an info log event log will be raised.

Severity

- Info
- Warning

Auto Clearing

No

Trigger Message

Status of a critical service has changed. Service Name: **{Critical Service Name}** , Service Type: **{Critical Service Type}** , Service Criticality: **{Critical Service Criticality Type}** , Service Instances: **{Critical Service Instances Count}** , Service Status: **{Critical Service Status}**

where:

- Critical Service Name is the Name of the critical service for which the status has changed
- Critical Service Type is the Type of the critical service namely Application, Common and Paas
- Critical Service Criticality Type is the Criticality type of the critical service namely Ready and Continuity
- Critical Service Instances Count is the Number of instances of a critical service which determine its availability/unavailability to the UPF instance
- Critical Service Status is the Status of a critical service namely up and down

Potential Impact

Change in status up/down might have direct impact on the UPF instance registration/deregistration respectively.

Repair Action

Check if all the required critical services are available for the UPF instance.

UPF Monitor Empty Active PPE List Event

Version

1.0

Description

This event indicates that the given UPF Monitor recieved empty Active PPE list from ETCD. Will trigger VNF Restart and deregistration in \$2 seconds if not restored.

Severity

- Critical

Auto Clearing

Yes

Clear Message

The UPF instance recieved non-empty active ppe list, will register with NRF now

Trigger Message

The UPF instance have empty active PPE List, will be deregistered with NRF. UPF Instance ID: **{NF Instance ID}**

where:

- NF Instance ID is the UUID of the UPF Intance

Potential Impact

UPF instance will not be able to provide its services in \$2 seconds due to which there may be temporary service discontinuity

Repair Action

Check if all the required configuration are correctly done and ETCD is UP.

UPF NFManagement Service Deregistration Event

Version

1.0

Description

This event indicates that the given UPF instance is deregistered with NRF due to either one of the critical services [Application/Common/Paas] or application configuration becomes unavailable due to which the UPF instance will not be able to provide its services.

Once the critical services or application configuration is restored resulting in the UPF instance registration with NRF, the event will be cleared.

Severity

- Critical

Auto Clearing

Yes

Clear Message

The UPF instance is in registered state with NRF now

Trigger Message

The UPF instance is deregistered with NRF. UPF Instance ID: **{NF Instance ID}**

where:

- NF Instance ID is the UUID of the UPF Instance

Potential Impact

UPF instance will not be able to provide its services due to which there may be temporary service discontinuity

Repair Action

Check if all the required critical services are available for the UPF instance along with its configuration

UPF NFManagement Service Deregistration Event Due to Application Status Change

Version

1.0

Description

This event indicates that the given UPF instance will be deregistered with NRF in 2 seconds due to application configuration becoming unavailable.

Once application configuration is restored the UPF instance registration with NRF will take place and the event will be cleared.

Severity

- Critical

Auto Clearing

Yes

Clear Message

The UPF instance is in registered state with NRF now since application status is restored.

Trigger Message

The UPF instance will be deregistered with NRF in ***{Application Deregistration Time Value}*** seconds because application status changed. UPF Instance ID: ***{NF Instance ID}***

where:

- NF Instance ID is the UUID of the UPF Instance
- Application Deregistration Time Value is the Duration in secs for which monitor for deregistration if application is down.

Potential Impact

UPF instance will not be able to provide its services due to which there may be temporary service discontinuity

Repair Action

Check if all the required critical services are available for the UPF instance along with its configuration

Unstable NF due to Frequent Restart

Version

1.0

Description

Network Function is unstable due to Frequent Restarts

Severity

- Critical

Auto Clearing

Yes

Clear Message

Critical Service:\$1 is stable.

Trigger Message

Due to Critical Service: **{Critical Service Name}** , NF: **{Network Function Name}** is unstable, timestamp: **{Event TimeStamp}**

where:

- Critical Service Name is the Critical Service Name
- Event TimeStamp is the Event Timestamp
- Network Function Name is the Network Function Name where:
- Critical Service Name is the Critical Service Name

Repair Action

Check why Critical Service instances are restarting

Alert Rules

Alert Rule 1

This event will result in raising an alert for every **Unstable NF due to Frequent Restart** event generation based on realert configuration. The rising of this alert is for each unique value of **severity**, **criticalServiceName** and **networkFunctionName** fields with realert time counted separately.

Unstable NF due to Readiness Failure

Version

1.0

Description

Network Function is unstable due to Readiness Failure

Severity

- Critical

Auto Clearing

Yes

Clear Message

Critical Service:\$1 is stable.

Trigger Message

Due to Critical Service: **{Critical Service Name}** , NF: **{Network Function Name}** is unstable, timestamp: **{Event TimeStamp}**

where:

- Critical Service Name is the Critical Service Name
- Event TimeStamp is the Event Timestamp
- Network Function Name is the Network Function Name where:
- Critical Service Name is the Critical Service Name

Repair Action

Check why there are Readiness Probe failures for the Critical Service

Alert Rules

Alert Rule 1

This event will result in raising an alert for every **Unstable NF due to Readiness Failure** event generation based on realert configuration. The rising of this alert is for each unique value of **severity**, **criticalServiceName** and **networkFunctionName** fields with realert time counted separately.